



UNIVERSITY OF THE PHILIPPINES

Master of Science in Materials Science and Engineering

Gian Carlo B. Carrillo

Investigation of a-Si/SiO₂ antireflective coatings for terahertz pump wavelengths

Thesis Adviser:

Arnel Salvador, Ph.D.

National Institute of Physics

University of the Philippines Diliman

TABLE OF CONTENTS

List of Tables	iv
List of Figures	v
List of Acronyms	vii
Chapter 1: Introduction	1
1.1 Significance of the study	2
1.2 Objectives of the study	2
1.3 Scope and limitations of the study	3
Chapter 2: Review of Related Literature	4
2.1 Terahertz photoconductive antenna	4
2.2 Electromagnetic Waves in Dielectric Materials	7
2.3 Transfer Matrix Method	10
2.4 Electron beam physical vapor deposition of a-Si and SiO ₂	12
2.5 Envelope method for refractive index	15
2.6 X-ray reflectivity	18
Chapter 3: Methodology	21
3.1 Initial growth of a-Si and SiO ₂ thin films via electron beam evaporation	21
3.2 Characterization of a-Si and SiO ₂ thin films	21

3.3	Simulation and growth of a-Si/SiO ₂ antireflective coatings	22
Chapter 4:	Results and Discussion	23
4.1	Thicknesses and densities of a-Si and SiO ₂ thin films	23
4.2	Refractive index of a-Si thin films	32
4.3	Refractive index of SiO ₂ thin films	34
4.4	Reflectance of a-Si/SiO ₂ antireflective coatings	36
Chapter 5:	Conclusion and Recommendations	39
References		40
Appendices		45
Chapter A:	Transfer Matrix Method python code	46

LIST OF TABLES

3.1	QCM inputs	22
4.1	a-Si thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra	23
4.2	SiO ₂ thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra	24
4.3	Thickness, roughness and densities calculated from XRR curve fitting .	24
4.4	Relevant values for the envelope method of the a-Si thin film.	33
4.5	Refractive indices and wavelengths for the SiO ₂ thin film	34
4.6	Simulated and experimental reflectance of the ARCs and Si wafer at different wavelengths	37

LIST OF FIGURES

2.1	THz spectra of substances	4
2.2	PCA geometries	5
2.3	THz TDS setup	6
2.4	Polarization at two interfaces	8
2.5	Single film on substrate	8
2.6	Propagation through different media	12
2.7	Structure zone model	15
2.8	Envelope method as described by Swanepoel	16
2.9	X-ray reflectivity	20
3.1	Normal incidence reflection spectroscopy	22
4.1	XRR measurements of a-Si thin films with varying QCM thicknesses .	26
4.1	XRR measurements of a-Si thin films with varying QCM thicknesses, continued	27
4.2	XRR measurements of SiO ₂ thin films with varying QCM thicknesses .	28
4.3	Linear regression of a-Si thicknesses	29
4.4	Linear regression of SiO ₂ thicknesses	29
4.5	XRR curve fit to 21.2 nm a-Si film	30
4.6	XRR curve fit to 27.5 nm a-Si film	30
4.7	XRR curve fit to 103.9 nm SiO ₂ film	30

4.8	XRR curve fit to 204 nm SiO ₂ film	31
4.9	Envelope method for 600 nm a-Si thin film	31
4.10	Refractive index of a-Si thin film	33
4.11	Reflectance of SiO ₂ thin film	35
4.12	Refractive index of SiO ₂ thin film	35
4.13	Reflectance of various ARCs	37

CHAPTER 1

INTRODUCTION

The development of devices which operate in the terahertz range has been a recent area of interest for researchers. The applications of terahertz waves include analysis of biological specimens [1], pharmaceuticals [2], and imaging [3] among others. The mechanisms by which these devices operate may include upconversion from the radio and mmWave range, or downconversion of photonic devices from the infrared region [4]. An example of a device which utilizes downconversion is the terahertz photoconductive antenna (PCA), and there are already multiple demonstrations and even commercial PCA devices that utilize the 780 nm and 1550 nm wavelengths, which are supplied by Ti:sapphire and telecommunication fiber lasers respectively [5], [6]. A PCA consists of a semiconductor material where the terahertz generation or absorption occurs and two metal electrodes on the semiconductor surface separated by a few microns that forms the photoconductive gap [7]. When a femtosecond pulse from the laser illuminates the gap, carriers are generated in the semiconductor and swept by an electric field from the bias applied to the metal electrodes, and the movement of the carriers generates the terahertz wave.

One of the common semiconductor materials of choice for a PCA is low temperature (LT) grown indium gallium arsenide (InGaAs), which has a bandgap of $E_g < 0.8$ eV and can be excited by common telecommunications wavelength lasers operating at 1550 nm [8]. Another common semiconductor material used for PCAs is LT gallium arsenide (GaAs), which has a bandgap of $E_g < 1.42$ eV and is operated at a much shorter wavelength of 780 nm[9]. However, when a bare PCA is illuminated by a laser there are significant reflection losses due to the high refractive index of the material[10]. Adding an antireflective coating to the PCA can increase the efficiency of the device by reducing the amount of light reflected by the film [11]. The challenge lies in the design

of a suitable antireflective coating for the aforementioned wavelengths.

1.1 Significance of the study

The development of antireflective coatings which reduce reflection losses at commonly used pump wavelengths for terahertz devices holds great potential for increasing the efficiency of terahertz (THz) PCA. By enhancing the efficiency of THz generation, this study contributes to the advancement of THz time domain spectroscopy (TDS) systems, which have diverse applications in fields such as biology, imaging, and pharmaceuticals. The Semiconductors group of the Condensed Matter Physics Laboratory in UP Diliman is able to grow LT-InGaAs and LT-InGaAs layers via molecular beam epitaxy (MBE), fabricate them into PCAs via photolithography, and study them using 1550 nm and 780 nm wavelength lasers. However, the PCAs being developed lack antireflective coatings optimized for their operating wavelengths. By investigating a multilayer antireflective coating (ARC) stack composed of a-Si and SiO₂, two materials commonly used for CMOS processing, this study aims to address these problems, improving the THz intensity generated by the PCA and its overall performance. The findings could pave the way for more efficient THz devices, enhancing capabilities in scientific research, medical diagnostics, and industrial processes.

1.2 Objectives of the study

The main objective of this study is to investigate the feasibility of using a-Si and SiO₂ thin films to create an antireflective coating which can be used for InGaAs and GaAs PCAs. Specifically, this will require the following objectives:

- to identify the optical properties of a-Si and SiO₂ thin films deposited using electron beam evaporation,
- to design an antireflective coating utilizing a-Si and SiO₂ optimized for the wavelengths at which LT-InGaAs and LT-GaAs THz PCAs can be operated, and

- to evaluate the reflectance of the fabricated a-Si/SiO₂ antireflective coating at the relevant wavelengths.

1.3 Scope and limitations of the study

The study will only cover the investigation of an amorphous silicon (a-Si) and SiO₂ ARC for the wavelengths at which LT-InGaAs and LT-GaAsTHz PCA are generally operated. As such, the study will not investigate the growth of LT-InGaAs and LT-GaAs film using MBE, nor the integration of the ARC with the photoconductive antennas, which will require photolithography and terahertz TDS. It will also be limited to non-reactive electron beam evaporation as the deposition method for the a-Si/SiO₂ and will not include investigation of other parameters during growth such as substrate heating and base pressure. The characterization is also limited to the evaluation of the reflectance of the designed coatings in the range of a Si photodetector (500 nm to 1000 nm).

CHAPTER 2

REVIEW OF RELATED LITERATURE

2.1 Terahertz photoconductive antenna

The range of frequencies between microwave and infrared in the electromagnetic spectrum is known as the THz range. It has also been called the THz gap previously due to the difficulty of finding emitters and detectors which worked in this frequency range[12]. THz radiation is non-ionizing and can penetrate through organic material such as plastic and clothing, thus it has potential use in biological characterization[1], detection of contraband materials[13], and chemical identification[14]. Some substances have specific THz absorption spectra which can be used to identify their composition, as shown in Figure 2.1. The interaction amongst the bonds and forces in the molecules result in vibrational modes visible in the THz region[13]. On the other hand, THz reflection can be used to detect concealed objects when the concealing material is transparent to THz radiation, as the case for dielectrics such as clothing[15]. Another common applications for THz is time-domain spectroscopy or TDS.

In THz TDS, a THz pulse is generated using a PCA. The shape of the electrodes influences the pattern of the THz radiation generated by the PCA. Dipole and bowtie

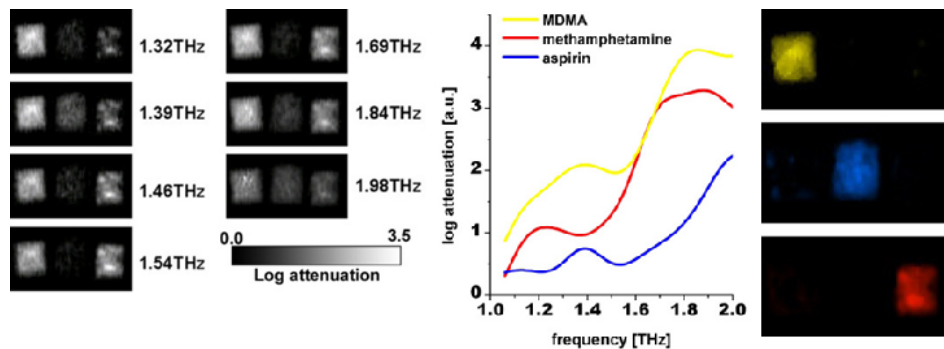


Figure 2.1: THz identification of MDMA, methamphetamine, and aspirin through a paper envelope. The substances are differentiated using their THz spectra[13]

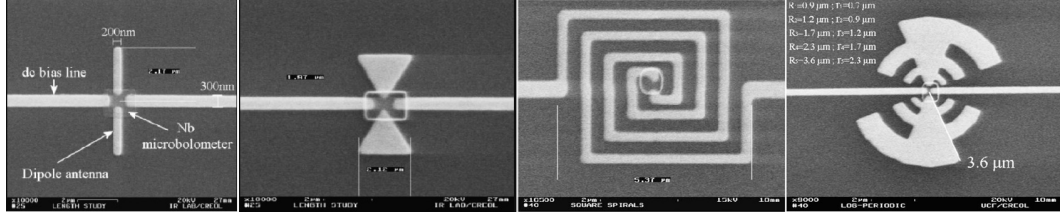


Figure 2.2: Commonly used THz PCA geometries; from left to right: dipole, bowtie, spiral, and log-spiral[16]

patterns demonstrate higher polarization ratios compared to spiral antennas while spiral and log-spiral antennas have a higher calculated radiation efficiency compared to bowtie and dipole antennas[16]. The shapes of the antennas are presented in Figure 2.2.

The emission mechanism for the PCA is as follows: the photoconductive material between the electrodes is illuminated with a femtosecond laser pulse, which generates photocarriers that are swept by an electric field applied to the electrodes of the PCA[7], [8]. The transient current $i_{PC}(t)$ resulting from the moving carriers generates electromagnetic radiation according to Maxwell's equations, with the intensity of the electric field component of the THz wave radiated by a dipole antenna expressed by Equation 2.1

$$E_{THz}(r, \theta, t) = \frac{l_e}{4\pi\epsilon c^2 r} \frac{\partial i_{PC}(t)}{\partial t} \sin(\theta) \quad (2.1)$$

where l_e is the antenna length, ϵ the permittivity, c the speed of light, r the distance from the dipole, and θ the angle relative to the dipole axis[7].

The transient current density term $i_{PC}(t)$ is proportional to the convolution integral of the carrier density $n(t)$ and electron velocity $v_e(t)$ as shown in Equation 2.2[7].

$$i_{PC}(t) \propto P_{pulse}(t) \otimes (n(t) v_e(t)). \quad (2.2)$$

The carrier density is described by

$$\frac{dn}{dt} = \eta \frac{P_{pulse}(t)}{h\nu} - \frac{n}{\tau_c} \quad (2.3)$$

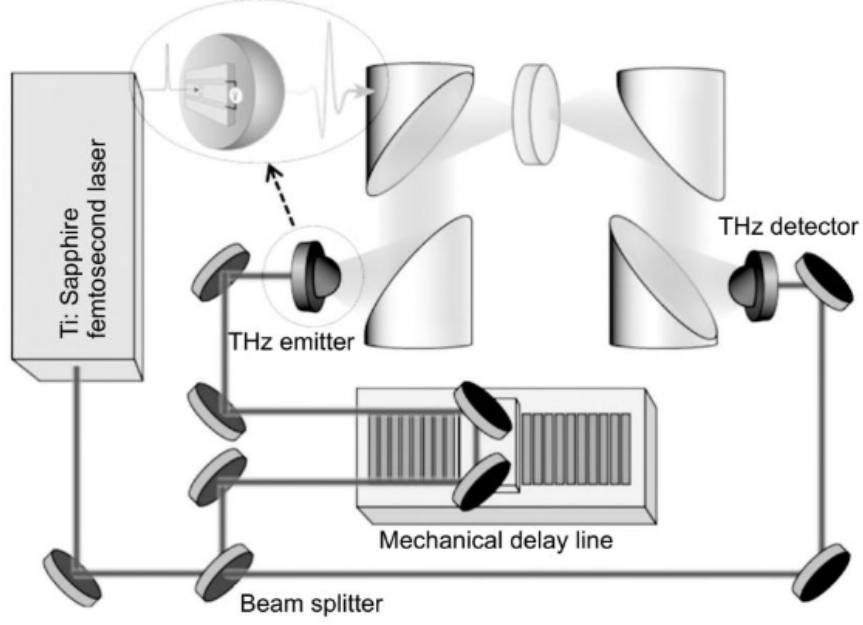


Figure 2.3: Schematic of a THz TDS setup with a femtosecond laser and two photoconductive antennas[7]

and using a 1D Drude-Lorentz model, the electron velocity by

$$\frac{dv_e}{dt} = -\frac{v_e}{\tau_s} + \frac{e}{m^*}E \quad (2.4)$$

where η is the absorption efficiency, $P_{\text{pulse}}(t)$ the ultrashort laser pulse power considered over time, τ_c , τ_s , and m^* the carrier lifetime, momentum relaxation time, electron effective mass respectively, and E the local electric field[7].

A photoconductive antenna is also used to detect THz radiation, with the incoming THz pulse focused together with a femtosecond pulse split from the same source at the gap of the antenna. As the femtosecond probe pulse arrives at the gap, carriers are excited and then accelerated proportionally by the electric field provided by the incoming THz pulse[8]. The difference of the probe pulse and the THz pulse provides a measurement in the time domain for the transient of the THz pulse. A schematic of a complete THz TDS system with emitter and detector is shown in Figure 2.3. Some of the photoconductive materials used for THz TDS systems include LT grown GaAs and InGaAs[17], [18], [19]. LT-GaAs is grown via molecular beam epitaxy at 200 °C to 300 °C instead of the usual growth temperature around 600 °C[20].

2.2 Electromagnetic Waves in Dielectric Materials

The propagation of electromagnetic waves through dielectric materials can be described using the Fresnel equations, which can be derived from the boundary conditions that constrain the electric and magnetic field components arising as a direct result of Maxwell's equations [21], [22]. Suppose incident light approaches a boundary between two dielectric materials with refractive indices n_1 and n_2 as shown in Figure 2.4. The first case shows s-polarized light, where the electric field is perpendicular to the plane of incidence, while the second case shows p-polarized light, where the electric field is parallel to the plane of incidence. The incident and reflected angles are equal $\theta_i = \theta_r$, and the transmitted angle is given by Snell's law $n_1 \sin \theta_i = n_2 \sin \theta_t$. The boundary conditions require that the components of the electric and magnetic fields be continuous, thus for s-polarized light, the electric fields are given by $E_i + E_r = E_t$ or $1 + r = t$, where r and t are the reflected and transmitted amplitudes and the incident amplitude is normalized to 1. The y-component of the magnetic field is given by $H_i \cos \theta_i - H_r \cos \theta_r = H_t \cos \theta_t$, which can be rewritten as $n_1 \cos \theta_i(1 - r) = n_2 \cos \theta_t t$ using the relation $H = nE/c$. Solving for r gives

$$r_s = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t} \quad (2.5)$$

while solving for t gives

$$t_s = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_t}. \quad (2.6)$$

For p-polarized light, the magnetic fields are given by $H_i + H_r = H_t$ or $n_1(1 + r) = n_2 t$, thus the electric fields are given by $E_i \cos \theta_i - E_r \cos \theta_r = E_t \cos \theta_t$, which can be rewritten as $n_1(1 - r) \cos \theta_i = n_2 t \cos \theta_t$. Solving for r gives

$$r_p = \frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_2 \cos \theta_i + n_1 \cos \theta_t} \quad (2.7)$$

while solving for t gives

$$t_p = \frac{2n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t}. \quad (2.8)$$

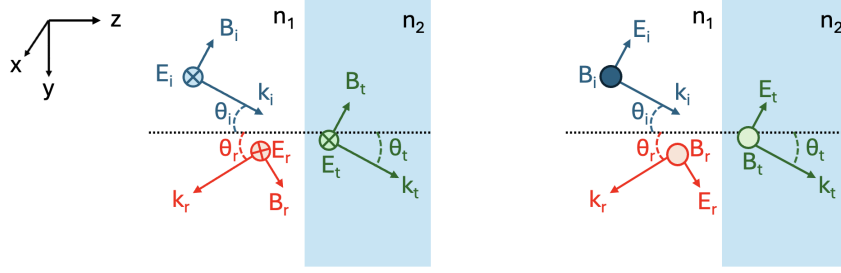


Figure 2.4: Diagram of polarized light at two interfaces. The left image shows the case for s-polarized light, with the electric field pointing into the page. For p-polarized light on the right the magnetic field points outside of the page.

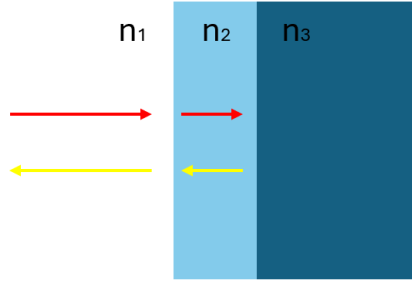


Figure 2.5: Diagram of light with normal incidence on a single film and substrate.

At normal incidence the reflectance R evaluates to

$$R = r_1^2 = \left| \frac{n_1 - n_2}{n_1 + n_2} \right|^2. \quad (2.9)$$

For an uncoated glass substrate ($n_{\text{glass}} \approx 1.5$) in air ($n_{\text{air}} \approx 1.0$), approximately 4% of incident light power is reflected per interface [23]. For high-index semiconductor materials like silicon or GaAs ($n_{\text{Si}} \approx 3.7$ at 780 nm), reflection losses exceed 30% [24].

Consider a single dielectric film of thickness d_2 and refractive index n_2 deposited between an ambient medium n_1 (air) and a substrate n_3 , where $n_1 < n_2 < n_3$ as shown in Figure 2.5. The reflectance of the structure at normal incidence would be given by

$$R = \frac{r_{12}^2 + r_{23}^2 + 2r_{12}r_{23} \cos(2\delta_2)}{1 + r_{12}^2 + r_{23}^2 + 2r_{12}r_{23} \cos(2\delta_2)} \quad (2.10)$$

where $\delta_2 = k_2 d_2 = 2\pi n_2 d_2 / \lambda$ is the phase difference between the reflected light at the interfaces of the dielectric film. The $2r_{12}r_{23} \cos(2\delta_2)$ term oscillates, and at $\cos(2\delta_2) = \pm 1$ or when $2\delta_2$ is an integer multiple of 2π , interference of the reflected waves occur

and the reflectance is at an extremum. The wavelength at the extremum is

$$\lambda = \frac{2n_2d_2}{m} \quad (2.11)$$

where m is the number of cycles from one extremum to another. The refractive index of the thin film can be calculated if the thickness is known

$$n_2 = \frac{m\lambda_1\lambda_m}{2d_2(\lambda_m - \lambda_1)}. \quad (2.12)$$

and the new λ is the average of the two wavelengths $\lambda = (\lambda_m + \lambda_1)/2$.

The mechanism of classical thin-film ARCs relies on destructive wave interference. Two necessary conditions must be satisfied simultaneously to achieve zero reflectance ($R = 0$) at a specific target wavelength λ_t [25]. The first condition is that the optical path difference between the partial wave reflected from the top interface (r_1) and the partial wave reflected from the substrate interface (r_2) must result in a π phase shift (180° out of phase). The physical film thickness d must satisfy:

$$d = \frac{\lambda_t}{4n_2}. \quad (2.13)$$

Since both reflection events occur at interfaces transitioning from a lower to a higher refractive index ($n_1 < n_2$ and $n_2 < n_3$), both undergo an phase change upon reflection. The round-trip optical path within the coating $2n_2d_2 = \lambda_t/2$ introduces a net phase change of π , ensuring destructive interference.

The second condition for zero reflectance is that the intensities of the two reflected waves must be identical $|r_1| = |r_2|$. Equalizing the Fresnel reflection coefficients yields:

$$\frac{n_2 - n_1}{n_2 + n_1} = \frac{n_3 - n_2}{n_3 + n_2} \implies n_2 = \sqrt{n_1 n_3}. \quad (2.14)$$

For a glass substrate ($n_3 = 1.5$) in air ($n_1 = 1.0$), the ideal coating index is $n_2 \approx 1.22$ [23]. However, finding natural bulk dielectric materials with $n \approx 1.22$ is difficult; magnesium fluoride (MgF_2 , $n \approx 1.38$) is usually employed as an alterna-

tive [26]. Single-layer ARCs suffer from multiple limitations [27]. First, destructive interference is exact only at the design wavelength λ_t , with the reflection increasing rapidly as the wavelength gets further away. Another limitation of single-layer ARCs is that off-normal angles of incidence alter both the internal propagation path length and the polarization-dependent reflection coefficients, shifting the minimum reflection away from λ_t .

To achieve broadband and wide-angle antireflection, multi-layer dielectric stacks comprising alternating high-index and low-index thin films are used. Multi-layer structures introduce multiple internal reflection pathways that result in broader and more easily tailored destructive interference minima across the relevant spectrum. The design of multi-layer ARCs is usually performed computationally with the transfer matrix method[28].

2.3 Transfer Matrix Method

At the boundary between two dielectric materials, the electric field amplitudes of the incident, reflected, and transmitted waves can be related using a transfer matrix [29], [30]. In the schematic shown in Figure 2.6a, a wave with amplitude E_1^+ is moving to the right, and another wave with amplitude E_1^- is moving to the left. For the other side of the boundary, the amplitudes are E_2^+ and E_2^- . The fields can be characterized using two 2×1 matrices, one for each side of the boundary, and a 2×2 interface matrix $\mathbf{D}$ as shown:

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} D_{12} & D_{21} \\ D_{12} & D_{21} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \quad (2.15)$$

The values of the matrix elements D_{ij} can be obtained from inspecting Figure 2.6a. The subscripts i and j refer to the direction of the wave propagation, with t_{12} signifying transmission from medium 1 to medium 2, and r_{12} signifying reflection from medium 1 to medium 2. The fields are thus given by

$$E_2^+ = t_{12}E_1^+ + r_{21}E_2^- \quad (2.16)$$

$$E_1^- = r_{12}E_1^+ + t_{21}E_2^- \quad (2.17)$$

which can be rearranged to

$$E_1^+ = \frac{E_2^+ - r_{21}E_2^-}{t_{12}} \quad (2.18)$$

$$E_1^- = \frac{r_{12}}{t_{12}} (E_2^+ - r_{21}E_2^-) + t_{21}E_2^- \quad (2.19)$$

Considering that the reflection coefficient is $r = (n_1 - n_2) / (n_1 + n_2)$ then $r_{12} = -r_{21}$. For the transmission coefficient $t = 2n_1 / (n_1 + n_2)$, and $t_{12} = n_2 t_{21} / n_1$. Substituting these relations into Equation 2.18 and Equation 2.19 with the assumption that $R + T = 1$ yields the matrix equation for the interface **D**

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \frac{1}{t_{12}} \begin{bmatrix} 1 & r_{12} \\ r_{12} & 1 \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \quad (2.20)$$

The propagation matrix **P** for a layer of thickness d and refractive index n can be obtained by considering the geometry of the waves in Figure 2.6b, where a layer with refractive index n and thickness d has been inserted between the two waves. The wave moving to the right varies as $E^+ = E_0 e^{-j(k_y y + k_z z - \omega t)}$, and the difference between the two waves at the left and right boundaries is given by $E_2^+ = E_1^+ e^{j k z \cos \theta}$ with the reverse given by $E_2^- = E_1^- e^{-j k z \cos \theta}$. The propagation matrix is thus given by

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \begin{bmatrix} e^{j k d \cos \theta} & 0 \\ 0 & e^{-j k d \cos \theta} \end{bmatrix} \begin{bmatrix} E_2^+ \\ E_2^- \end{bmatrix} \quad (2.21)$$

where $k = 2\pi n / \lambda$ is the wavenumber in the medium. This analysis can be extended to a two layer system by multiplying the interface and propagation matrices while considering the order of the layers. Starting from the rightmost boundary as shown in Figure 2.6c, the fields on both sides are related using the interface matrix by $\mathbf{E}_{3R} = \mathbf{D}_{34} \mathbf{E}_4$, where $3R$ is the field at the right side of layer 3. The fields at the left side of layer 3

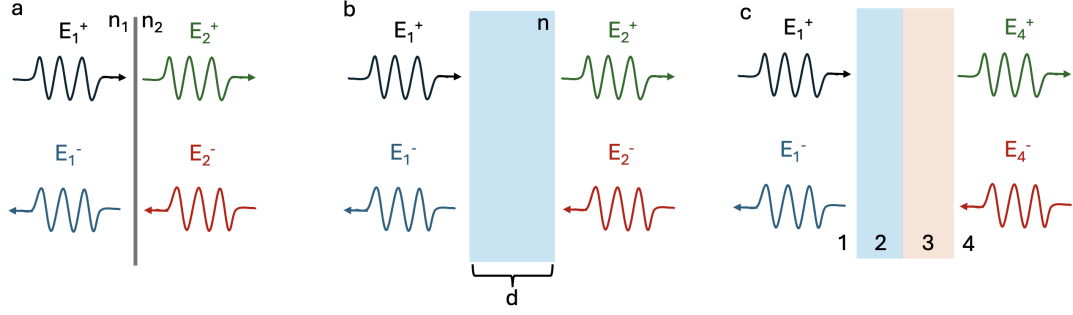


Figure 2.6: Propagation of waves at an interface (a), through a layer with refractive index n and thickness d (b), and through a multilayer system (c).

are related to the right side by the propagation matrix $\mathbf{E}_{3L} = \mathbf{P}_3 \mathbf{E}_{3R}$, which in turn can be related using another interface matrix to the field across the relevant boundary. This results in the following matrix equation relating the fields at the leftmost region to the rightmost region:

$$\mathbf{M} = \mathbf{D}_{12} \mathbf{P}_2 \mathbf{D}_{23} \mathbf{P}_3 \mathbf{D}_{34} \quad (2.22)$$

such that

$$\begin{bmatrix} E_1^+ \\ E_1^- \end{bmatrix} = \mathbf{M} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} E_4^+ \\ E_4^- \end{bmatrix}. \quad (2.23)$$

This method allows the analysis of any multilayer system using matrix multiplication of the component interface $\mathbf{D}$ and propagation $\mathbf{P}$ matrices. The reflection coefficient r for a multilayer structure is simply

$$r = \frac{M_{21}}{M_{11}} \quad (2.24)$$

since $t = 1/M_{11}$ and $r = M_{21}/M_{11}$.

2.4 Electron beam physical vapor deposition of a-Si and SiO₂

Silicon and its native oxide SiO₂ are the most common materials used in the semiconductor industry. Thin films of Si and SiO₂ are used in microelectronics, photonics, and

photovoltaic applications [31]. These thin films are grown using methods ranging from physical vapor deposition, chemical vapor deposition, to thermal oxidation. Electron beam physical vapor deposition (EBPVD) or e-beam evaporation is one commonly utilized technique for depositing high-purity Si and SiO₂ films due to its high deposition rate, precise thickness control, and capability to vaporize high-melting-point materials [32].

SiO₂ is widely utilized due to its wide bandgap of 8.9 eV, low refractive index of $n \approx 1.45 - 1.46$ in the visible spectrum, high chemical stability, and low dielectric loss for optical coatings [33]. While thermal oxidation can produce high-quality native SiO₂ from Si, it requires temperatures $>900^\circ\text{C}$ and is restricted to silicon substrates [34]. Electron beam evaporation enables low-temperature deposition $<300^\circ\text{C}$ on different substrates including polymers, glass, and compound semiconductors such as GaAs [35]. However, e-beam evaporation of SiO₂ presents unique physical challenges including dissociation during evaporation, non-stoichiometric SiO₂, porosity, and intrinsic film stress [36].

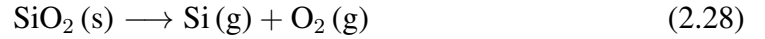
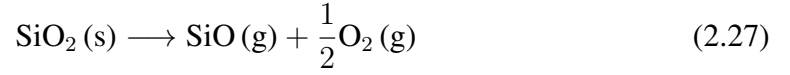
In e-beam evaporation, a high-energy electron beam is magnetically steered onto a high-purity target contained in a water-cooled crucible [32]. The electron generation is caused by Richardson-Dushman thermionic emission from a heated filament which is typically tungsten:

$$J = AT^2 \exp\left(-\frac{\Phi}{k_B T}\right) \quad (2.25)$$

where J is the emission current density A/m², $A \approx 1.2 \times 10^6 \text{ A/(m}^2 \text{ K}^2)$ is the Richardson constant, Φ is the work function of the filament material in eV, k_B is the Boltzmann constant, and T is the absolute temperature in K. Emitted electrons are accelerated toward an anode using a high potential difference and the beam trajectory is controlled via the Lorentz force:

$$\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2.26)$$

In a uniform transverse magnetic field B , electrons follow a circular path. This bending caused by the magnetic field allows the electron gun to remain shielded from direct line-of-sight deposition from the source while enabling frequent sweeping across the crucible to maintain uniform evaporation from the source as shown in Figure 2.4. As the electron beam strikes the source, local temperatures at the beam spot may exceed 1700°C , causing melting and evaporation for Si or direct sublimation in the case of SiO_2 [37]. Unlike pure Si, SiO_2 does not evaporate purely as intact SiO_2 molecules. Thermal dissociation may occur at the surface of the source, yielding a mixture of gaseous species:



Non-reactive e-beam evaporation of SiO_2 in high vacuum typically yields films of SiO_x with lower stoichiometric values $x < 2.0$. A lower stoichiometry impacts film performance by introducing sub-oxide states (Si^{1+} , Si^{2+} , Si^{3+}) and oxygen vacancies, leading to an increased optical absorption and increased refractive index. To ensure fully stoichiometric SiO_2 , reactive e-beam evaporation can be performed by introducing oxygen gas into the chamber.

According to the structure zone model, the growth of the films proceed in three general steps [38]. The first step occurs when the vaporized molecules are transported towards the substrate, the second when they are adsorbed onto the surface of the substrate or growing film, their diffusion across the surface, and any chemical reactions that may involve their incorporation into the film or removal via thermal desorption or sputtering. The third step involves the movement of the molecules to their final locations via bulk diffusion and solid state reactions.

Movchan and Demchishin classified film growth into distinct zones, with the morphology of the films depending on the normalized homologous temperature $T_h = T_{\text{sub}}/T_m$ where T_{sub} is the substrate temperature and T_m is the melting point of the material in K as shown in Figure 2.7 [39]. At low temperatures ($T_h < 0.3$) surface

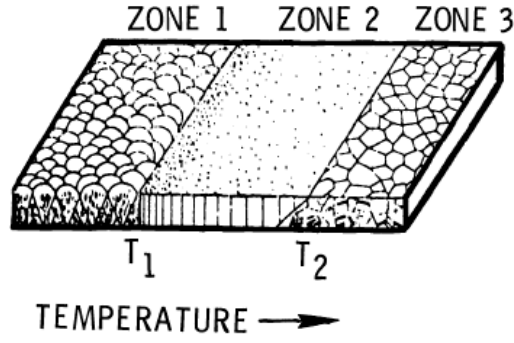


Figure 2.7: Structure zone diagram introduced by Movchan and Demchishin from study of vacuum evaporated metals and oxides.

diffusion is low, and molecules essentially stick where they arrive, producing porous, tapered columnar structures. This is considered as Zone 1. At Zone 2 ($T_h < 0.5$) the temperatures are high enough to allow the surface diffusion fluxes to dominate over the arrival fluxes, but still low enough such that the bulk diffusion rates remain orders of magnitude less than the surface diffusion rates. When the temperature is high enough, bulk diffusion will dominate over all other processes and the formation is characterized by fully dense grains. This is considered as Zone 3. The morphology of the films also has an effect on the refractive index, with porous films generally having a lower index compared to the bulk.

2.5 Envelope method for refractive index

Accurate characterization of the optical properties of weakly absorbing thin films deposited on transparent substrates is crucial for integrated optical devices [40]. Building upon earlier developed interference envelope concepts [41], Swanepoel formulated a method which relies solely on the upper (T_M) and lower (T_m) continuous envelope curves fitted to the fringe extrema of a single normal-incidence transmission spectrum $T(\lambda)$ as shown in Figure 2.8 [42]. Consider a homogeneous thin dielectric film of uniform physical thickness d , complex refractive index $\tilde{n} = n - ik$, deposited on a transparent, thick substrate of index s surrounded by air ($n_0 = 1$) as shown in Figure 2.5. In the weakly absorbing region ($k^2 \ll n^2$), internal multiple reflections give

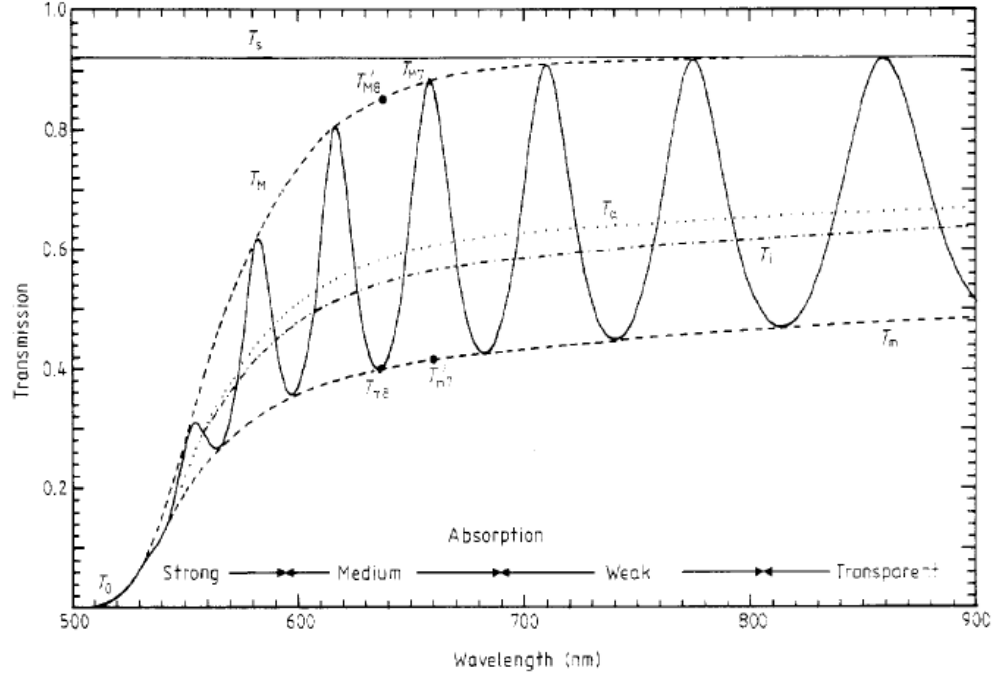


Figure 2.8: Envelope method as described by Swanepoel for a 1 μm film of a-Si:H on a finite glass substrate with transmittance T_s .

rise to alternating constructive and destructive interference fringes in the transmission spectrum $T(\lambda)$.

The optical transmittance T at normal incidence, accounting for incoherent reflections at the rear substrate boundary, is given by:

$$T = \frac{Ax}{B - Cx \cos(\phi) + Dx^2} \quad (2.29)$$

where: $x = \exp(-\alpha d)$ is the absorbance term, and $\alpha = 4\pi k/\lambda$ is the absorption coefficient, $\phi = 4\pi nd/\lambda$ is the phase shift per round trip in the film. The system constants A , B , C , and D depend strictly on n and s :

$$A = 16n^2s$$

$$B = (n + 1)^3(n + s^2)$$

$$C = 2(n^2 - 1)(n^2 - s^2)$$

$$D = (n - 1)^3(n - s^2)$$

where $s = 1/T_s + \sqrt{1/T_s - 1}$ is the refractive index of the substrate and T_s the transmittance of the substrate. The transmission spectrum exhibits local maxima T_M when at conditions of constructive interference $\cos(\phi) = +1$ and local minima T_m when $\cos(\phi) = -1$ or destructive interference. Continuous envelope curves $T_M(\lambda)$ and $T_m(\lambda)$ across the spectrum can be created using various functions, such as the McClain algorithm [43], parabolic or exponential interpolation [44].

For exponential interpolation, the envelope follows the following mathematical expression:

$$f^{\text{exp}}(\lambda, T_0, T_1, T_2, \Lambda_1, \Lambda_2) = T_0 + T_1 e^{-\lambda/\Lambda_1} + T_2 e^{-\lambda/\Lambda_2} \quad (2.30)$$

where the optimal parameters $(\lambda, T_0, T_1, T_2, \Lambda_1, \Lambda_2)$ are obtained by solving a nonlinear-least squares fitting problem. Although this method is less precise than other algorithms, it has the benefit of being more appropriate for transmission spectra with a limited number of oscillations and tangent points, which often occurs for ultra-thin films. Common interpolations typically require at least four tangent points for a reliable estimation of each envelope [43], [44].

By setting $\cos(\phi) = \pm 1$ in the fundamental transmittance equation, an explicit equation for the refractive index n at any continuous wavelength λ in the transparent/weakly absorbing regime ($x \approx 1$) can be derived [42]:

$$n = \left[N + \sqrt{N^2 - s^2} \right]^{1/2} \quad (2.31)$$

where

$$N = 2s \left(\frac{T_M - T_m}{T_M T_m} \right) + \frac{s^2 + 1}{2} \quad (2.32)$$

From the interference condition, the fundamental equation for fringe order m is:

$$2nd = m\lambda \quad (2.33)$$

where m is an integer for maxima (T_M) and a half-integer for minima (T_m). If n_a and n_b are the calculated refractive indices at two adjacent extrema wavelengths λ_a and λ_b

then thickness can be determined:

$$d = \frac{\lambda_a \lambda_b}{2(\lambda_a n_b - \lambda_b n_a)} \quad (2.34)$$

Taking the average over multiple adjacent fringe pairs yields a more accurate value for physical thickness d [42], [45]. Once an initial value of d is obtained, exact integer and half-integer values of m_0 are determined by evaluating Equation 2.33. Fitting these rounded integers back into Equation 2.33 gives more refined values of n and d for the extrema. The resulting values of $n(\lambda)$ can be fitted to Sellmeier's equation to obtain a continuous function of n with the wavelength:

$$n^2(\lambda) = 1 + \frac{B_1 \lambda^2}{\lambda^2 - C_1} + \frac{B_2 \lambda^2}{\lambda^2 - C_2} \quad (2.35)$$

2.6 X-ray reflectivity

As multi-layer optical stacks require nanometer precision, non-destructive surface characterization techniques are essential [46], [47]. X-ray reflectometry (XRR) is a non-destructive optical characterization technique widely employed to determine the structural properties of thin films and multi-layer heterostructures with nanometer resolution. By measuring the speculatively reflected intensity of X-rays as a function of incident angle, XRR provides absolute determinations of film thickness d , electron density ρ_e , and interface or surface roughness σ . For hard X-rays $\lambda \approx 0.05 - 0.2$ nm, the optical refractive index n of condensed matter is slightly less than unity:

$$n = 1 - \delta - i\beta \quad (2.36)$$

where δ represents the dispersion term, which governs phase velocity and wave deflection and β represents the absorption term, which governs wave attenuation.

In terms of microscopic atomic parameters, δ and β are directly linked to the bulk

properties of the material [47]:

$$\delta = \frac{r_e \lambda^2}{2\pi} \rho_e = \frac{r_e \lambda^2 N_A}{2\pi} \sum_j \frac{\rho_m (f_j^0 + f_j')}{M_j} \quad (2.37)$$

$$\beta = \frac{\mu \lambda}{4\pi} = \frac{r_e \lambda^2 N_A}{2\pi} \sum_j \frac{\rho_m f_j''}{M_j} \quad (2.38)$$

where $r_e = e^2/4\pi\epsilon_0 m_e c^2 \approx 2.818 \times 10^{-15}$ m is the classical electron radius, ρ_e is the electron density of the material, N_A is Avogadro's number, ρ_m is the mass density, and M_j is the atomic molar mass, $(f_j^0 + f_j' + i f_j'')$ represents the complex atomic scattering factor, and μ is the linear x-ray attenuation coefficient. Because δ is positive and typically on the order of 10^{-6} to 10^{-5} , the real part of n ($1 - \delta$) is less than 1.0. Consequently, vacuum or air behaves as an optically denser medium than any solid or liquid in the X-ray regime.

Because $n < 1$, X-rays impinging on a material surface from air/vacuum ($n_0 \approx 1$) undergo total external reflection when the incident grazing angle θ is below a critical angle θ_c . Applying Snell's Law ($\cos \theta_c = n = 1 - \delta$):

$$\cos \theta_c = 1 - \delta \quad (2.39)$$

Using the Taylor expansion $\cos \theta_c \approx 1 - \theta_c^2/2$ for small angles:

$$\theta_c \approx \sqrt{2\delta} = \lambda \sqrt{\frac{r_e \rho_e}{\pi}} \quad (2.40)$$

At angles $\theta \leq \theta_c$, 100% of the incident X-ray intensity is reflected ($R = 1$), and the electromagnetic wave inside the material decays exponentially as an evanescent wave with a penetration depth of only a few nanometers. Above θ_c , the reflectivity drops sharply, decaying as θ^{-4} according to Fresnel's law [46].

When an X-ray beam strikes a thin film deposited on a substrate of different electron density, waves reflect from both the top interface of air and film and the bottom interface of the film and substrate. Constructive and destructive interference between

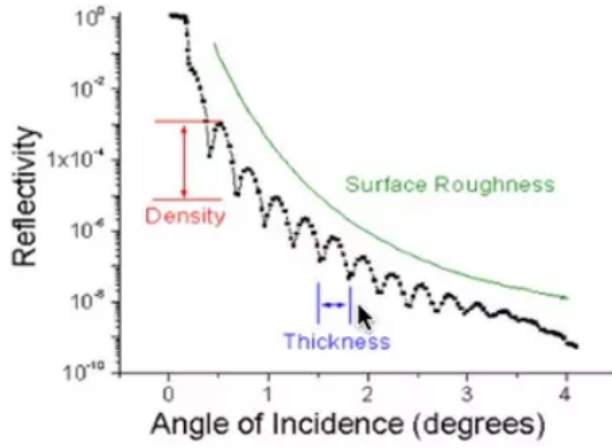


Figure 2.9: Effects of thickness, surface roughness and difference in density on the XRR intensity

these coherent wave paths produces periodic oscillations in reflectivity known as Kiessig fringes [48]. The fringe spacing $\Delta\theta$ is inversely proportional to the film thickness d :

$$d \approx \frac{\lambda}{2\Delta\theta}. \quad (2.41)$$

Real surfaces exhibit physical roughness (σ) and interdiffusion across boundaries. Roughness scatters X-rays non-specularly, dampening the reflected amplitude. The reduction in specular reflectivity due to root-mean-square (RMS) roughness σ is modeled using the Nevot-Croce factor [49], which modifies the ideal Fresnel reflection coefficient $r_{j,j+1}$:

$$r_{j,j+1}^{\text{rough}} = r_{j,j+1}^{\text{ideal}} \exp(-2k_{z,j}k_{z,j+1}\sigma_{j,j+1}^2) \quad (2.42)$$

where $k_{z,j} = 2\pi n_j \sin \theta_j / \lambda$ is the z -component of the wave vector in layer j . Increased roughness causes a faster drop-off in total reflectivity and dampens Kiessig fringe amplitudes. The effect of the three parameters on the XRR intensity is shown in Figure 2.9.

CHAPTER 3

METHODOLOGY

3.1 Initial growth of a-Si and SiO₂ thin films via electron beam evaporation

Glass slides and Si wafers were used as substrates for the initial growth via electron beam evaporation of a-Si and SiO₂ thin films respectively. The substrates were cleaned by sonication in acetone, isopropanol, and deionized water for 5 minutes each, after which they were blow-dried with nitrogen gas. The substrates were then loaded into the ULVAC electron beam evaporation chamber, where a base pressure of 4.5×10^{-4} Pa was achieved before actual deposition. Deposition was carried out at room temperature, and the nominal deposition rates and thicknesses of the films were monitored using a quartz crystal microbalance (QCM) with the input values in Table 3.1.

3.2 Characterization of a-Si and SiO₂ thin films

After unloading the samples from the chamber, the thicknesses of the films were measured using x-ray reflectivity (XRR) with a Malvern Panalytical Empyrean x-ray diffractometer. Curve fitting was performed on selected samples to obtain the densities and interface roughness of the films.

The refractive index of the a-Si film was determined using Swanepoel's method via UV-Vis spectroscopy with a Shimadzu UV-Vis spectrophotometer while the refractive index of the SiO₂ film was identified using the interference method via the reflectance spectroscopy setup as shown in Figure 3.1. The measured refractive indices of the a-Si and SiO₂ films were used for the design of the a-Si/SiO₂ ARC stack.

	z-factor	ρ g/cm ³
Si	0.712	2.32
SiO ₂	1	2.65

Table 3.1: QCM inputs

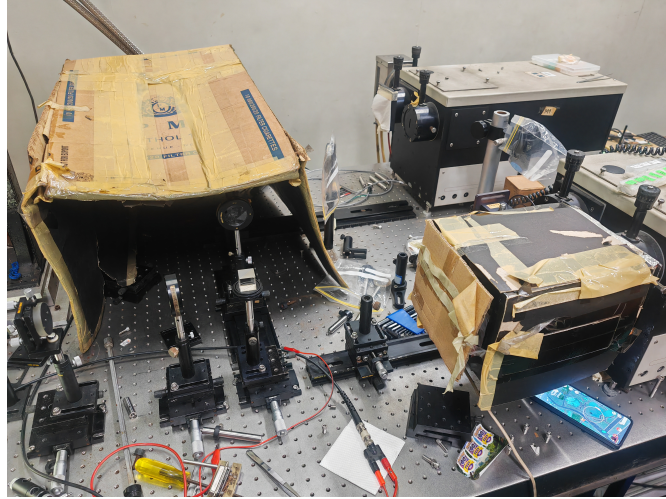


Figure 3.1: Normal incidence reflectance spectroscopy setup.

3.3 Simulation and growth of a-Si/SiO₂ antireflective coatings

A transfer matrix method (TMM) implementation in Python was used to calculate the optimal thicknesses of the Si and SiO₂ layers for the antireflective coating stack. The refractive indices of the a-Si and SiO₂ layers used for the simulation were obtained from the previous section. The optimal thicknesses of the layers were calculated to minimize the reflectance at 1550 nm and 780 nm, which are the wavelengths of the lasers which can be used in the THz-TDS spectroscopy system. The program iterated over a range of thicknesses for each layer while calculating the reflectance of the stack, selecting the thicknesses that resulted in the lowest reflectance at the target wavelengths. The thicknesses obtained from the simulation were then used for the growth of the a-Si/SiO₂ ARC on bare wafers via electron beam evaporation. The reflectance of the samples with the antireflective coating was measured via reflectance spectroscopy and compared to the simulated reflectance to verify the accuracy of the simulation and growth process.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Thicknesses and densities of a-Si and SiO₂ thin films

Figure 4.1 and Figure 4.2 show the XRR measurements of various a-Si films grown on glass substrates and SiO₂ films grown on Si(100) substrates respectively with varying thicknesses according to the QCM readings across different growth runs. From the graphs of the intensity the 2θ range at which the detector can count the reflected x-rays generally ranges from 0.3° to 2.0° . The measured thicknesses of the a-Si film based on XRR are summarized in Table 4.1, while for SiO₂ the results are summarized in Table 4.2. No thickness data could be obtained from XRR for films which were thicker than $1.52 \text{ k}\text{\AA}$ for a-Si and $1.369 \text{ k}\text{\AA}$ for SiO₂.

Figure 4.5, Figure 4.6, Figure 4.7 and Figure 4.8 show the simulated curves using the XRR software for selected a-Si films on glass substrates and SiO₂ films on Si. The values of thickness, interface roughness and densities obtained by fitting the curve are summarized in Table 4.3.

Linear regression of the measured thicknesses versus the QCM readings for a-Si as shown in Figure 4.3 yields the equation

QCM (k $\text{\AA}$)	Direct (nm)	Fourier (nm)
0.035	7.822	6.90
0.069	14.077	13.80
0.104	21.290	21.20
0.138	27.529	27.50
1.270	214.620	213.15
1.520	259.840	259.20

Table 4.1: a-Si thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra

QCM (kÅ)	Direct (nm)	Fourier (nm)
0.525	104.6	103.9
0.884	165.1	163.2
1.050	206.6	204.7
1.369	272.3	270.1

Table 4.2: SiO₂ thicknesses measured using the QCM, direct fringe calculation and Fourier method from the XRR spectra

	Thickness (nm)	Roughness (nm)	Density (g/cm ³)
21.20 nm a-Si	21.9115	0.7047	1.80726
27.50 nm a-Si	28.4339	0.8008	1.75748
103.9 nm SiO ₂	104.5311	0.3626	1.69738
204.7 nm SiO ₂	203.8994	0.741	2.04566

Table 4.3: Thickness, roughness and densities calculated from XRR curve fitting

$$y = 168.03x + 3.0403 \quad (4.1)$$

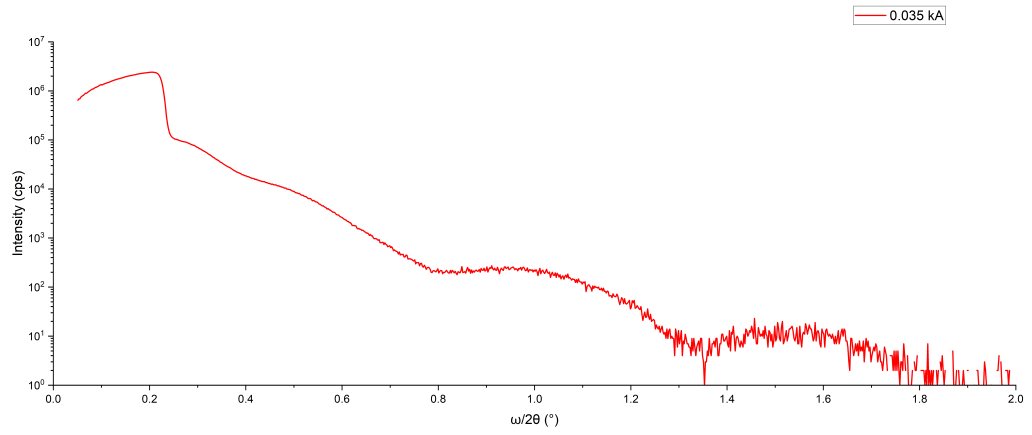
with $R^2 = 0.9999$, where y is the measured XRR thickness in nm and x is the QCM reading in kÅ. Similarly, linear regression of the measured XRR thickness and the QCM values for SiO₂ as shown in Figure 4.4 yields the following equation with $R^2 = 0.9941$:

$$y = 198.52x - 4.5056 \quad (4.2)$$

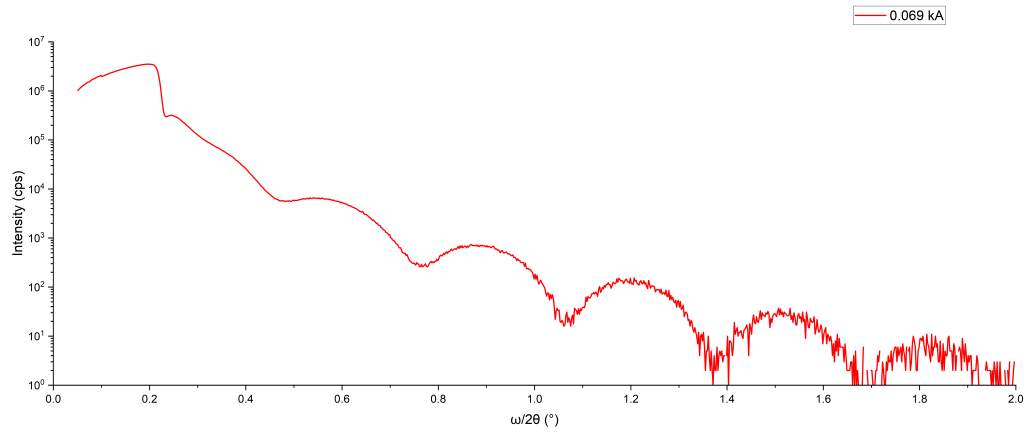
The initial comparison of the QCM readings and the measured thicknesses during XRR indicate that the QCM readings during deposition generally underestimate the actual thicknesses of the films. This underestimation is likely due to the mismatch in densities used as input for the QCM as shown in Table 3.1 and the actual densities of the evaporated film as shown in Table 4.3. The thicknesses of the films as obtained by curve fitting generally agree with the direct and Fourier method of analysis for the Kiessig fringes. The low interface roughness of the films imply that electron beam evaporation is an effective method to produce uniform films, albeit with much lower densities than the bulk material. This discrepancy may be caused by the increased porosity of the

film. The lack of data for films thicker than 1.52 kÅ for a-Si and 1.369 kÅ for SiO₂ is explained by the limitation of the equipment, which can only resolve fringes larger than that which correspond to a thickness of 270 nm.

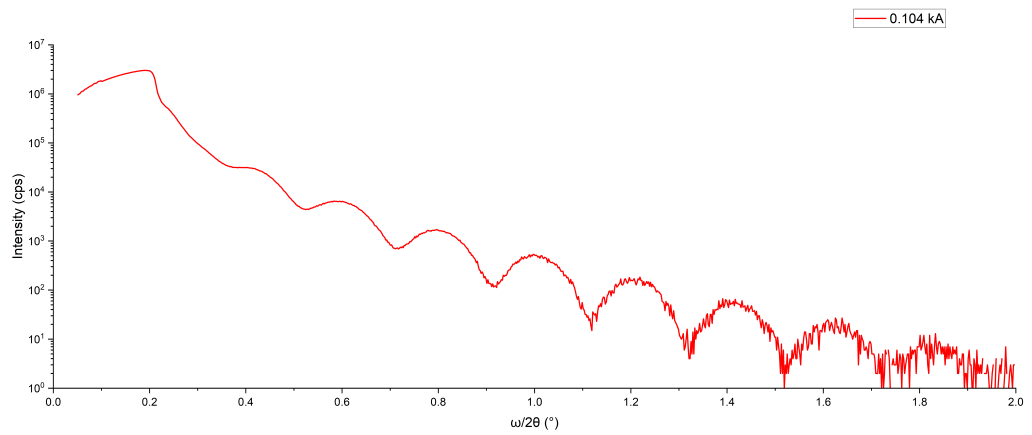
Nevertheless, the results of the regression analysis indicate that the QCM readings scale linearly with the actual thicknesses of the films as measured by XRR. The high R^2 values suggest that the QCM is a reliable tool for monitoring film thickness during deposition as long as the linear relationship is considered, and that the linear relationships obtained in this study can be used to calibrate the QCM readings for future depositions. The lower R^2 value in the regression for SiO₂ is attributed to the variation in the deposition rate during the growth of the SiO₂ films, which is caused by the difficulty in maintaining a stable deposition rate for SiO₂ due to its low thermal conductivity and tendency to sublime instead of melting into a uniform liquid pool during heating by the electron beam as is the case for other materials.



(a) 0.035 kÅ a-Si thin film

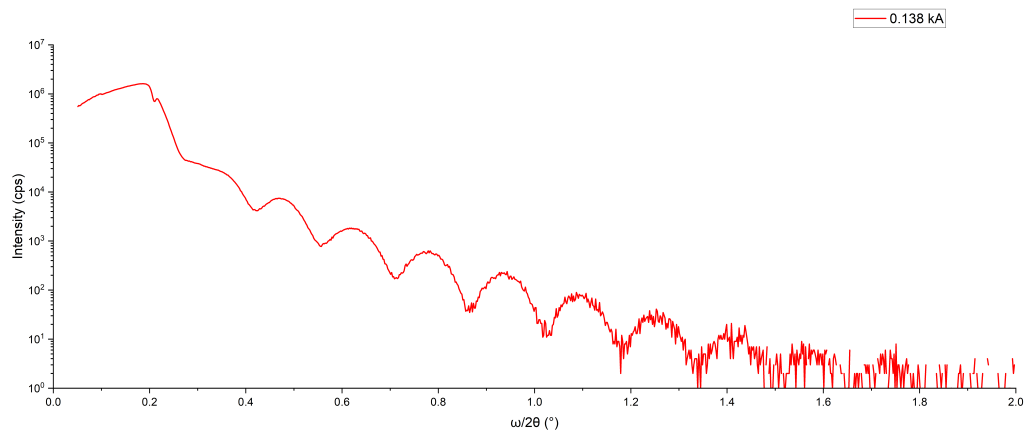


(b) 0.069 kÅ a-Si thin film

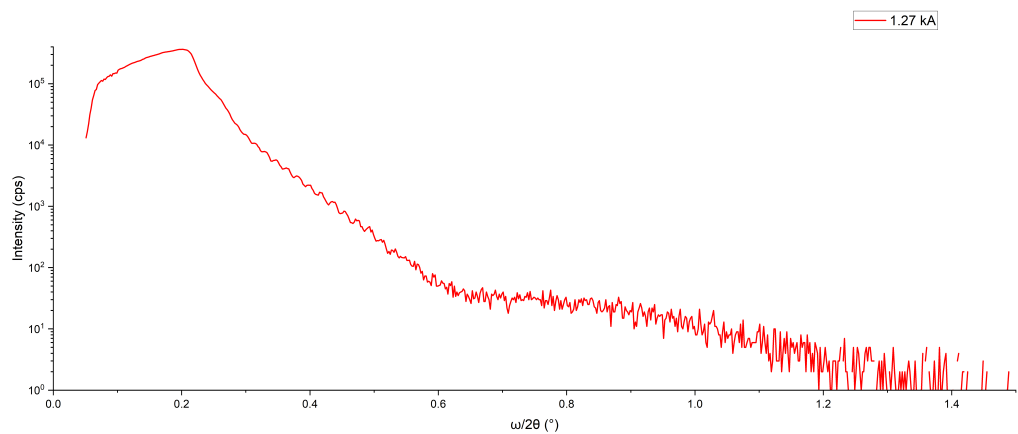


(c) 0.104 kÅ a-Si thin film

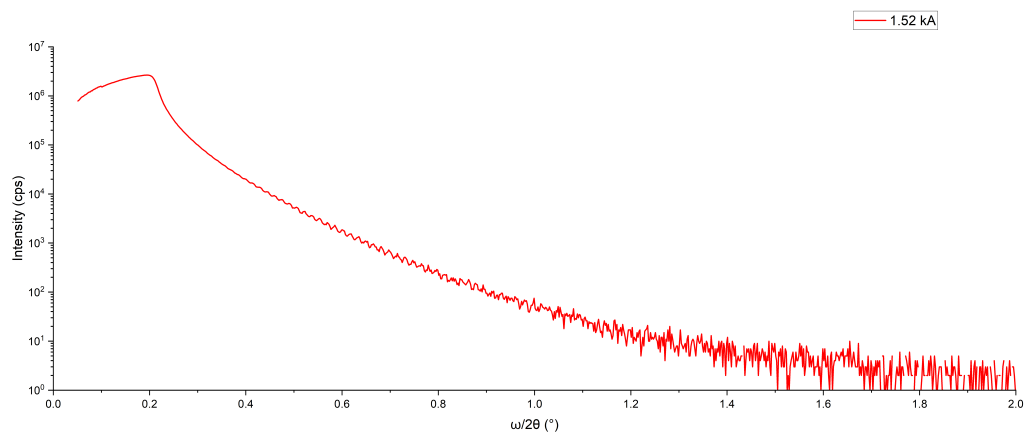
Figure 4.1: XRR measurements of a-Si thin films with varying QCM thicknesses



(d) 0.138 kÅ a-Si thin film

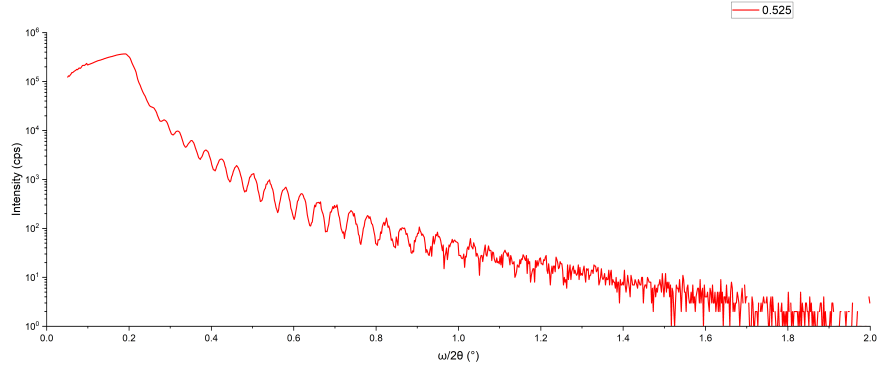


(e) 1.270 kÅ a-Si thin film

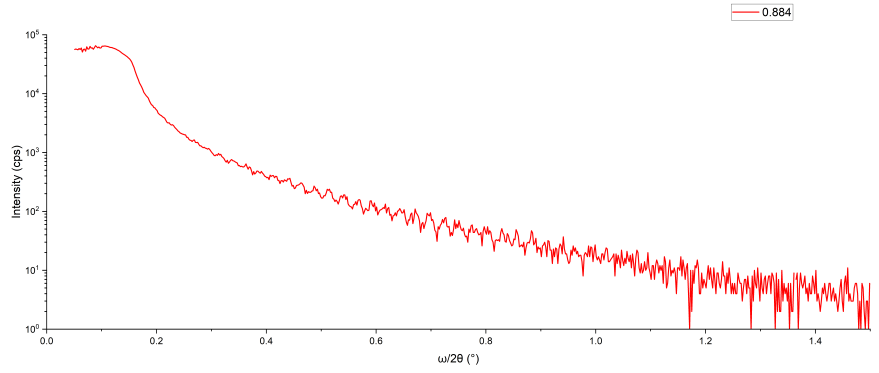


(f) 1.520 kÅ a-Si thin film

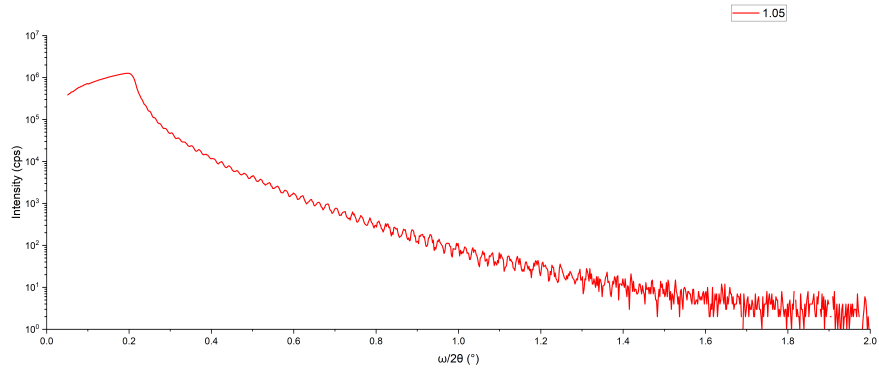
Figure 4.1: XRR measurements of a-Si thin films with varying QCM thicknesses, continued



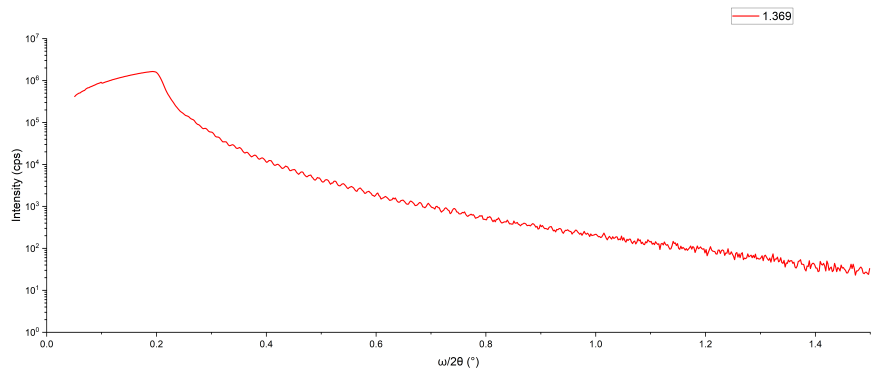
(a) 0.525 kÅ SiO₂ thin film



(b) 0.884 kÅ SiO₂ thin film



(c) 1.050 kÅ SiO₂ thin film



(d) 1.369 kÅ SiO₂ thin film

Figure 4.2: XRR measurements of SiO₂ thin films with varying QCM thicknesses

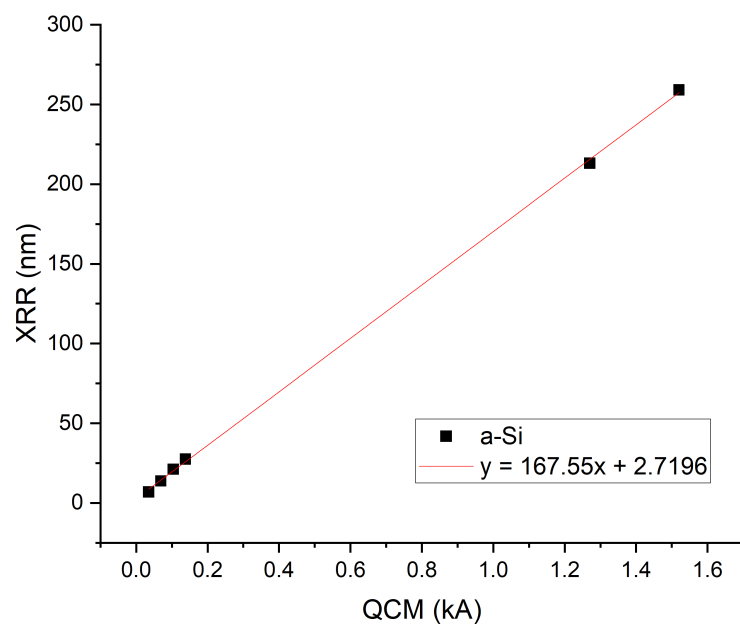


Figure 4.3: Linear regression of XRR and QCM thicknesses for a-Si thin films grown via e-beam evaporation.

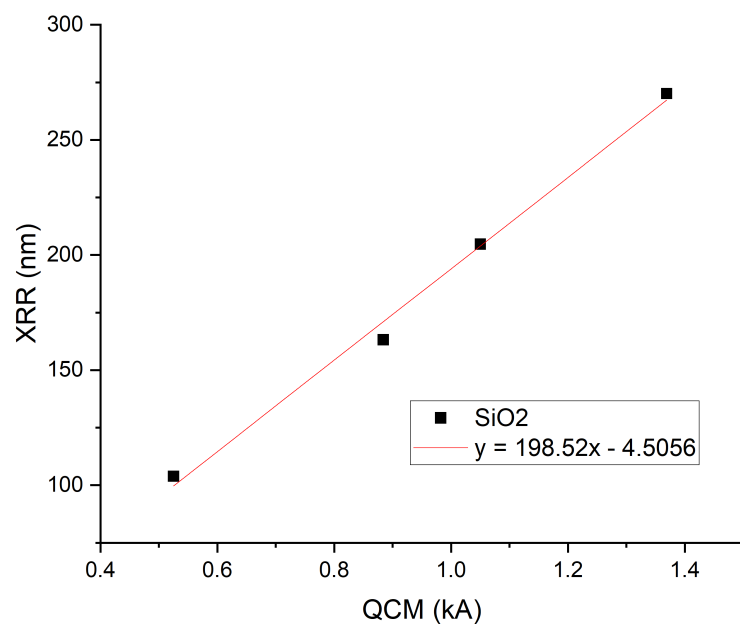


Figure 4.4: Linear regression of XRR and QCM thicknesses for SiO₂ thin films grown via e-beam evaporation.

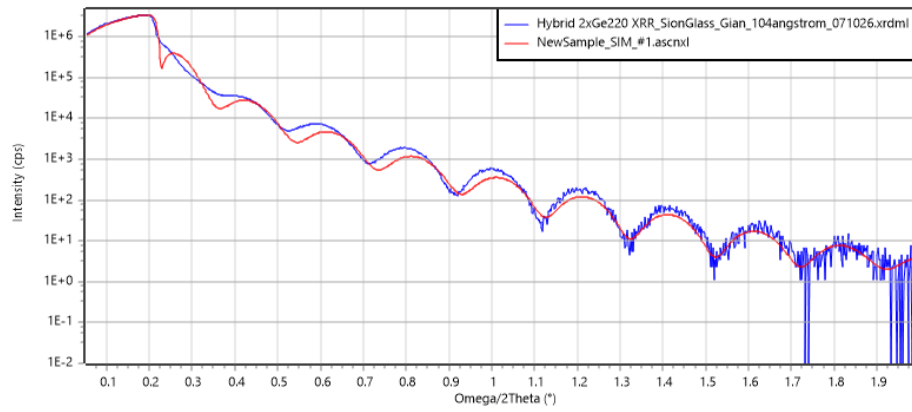


Figure 4.5: XRR curve fit to 21.20 nm a-Si thin film

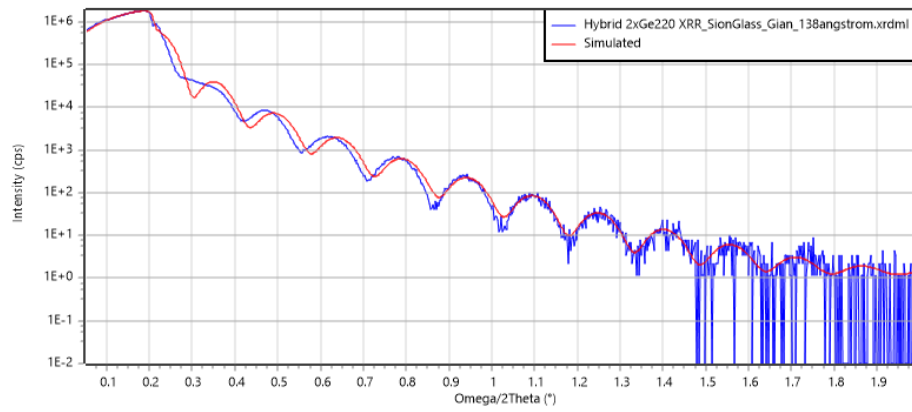


Figure 4.6: XRR curve fit to 27.50 nm a-Si thin film

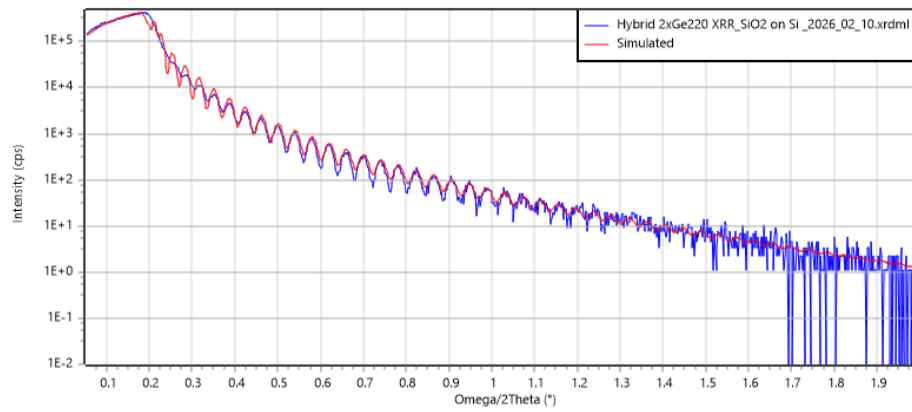


Figure 4.7: XRR curve fit to 103.9 nm SiO₂ thin film

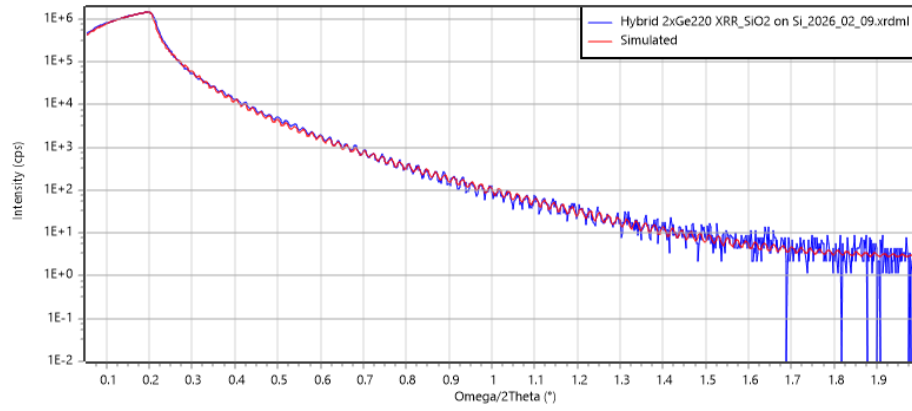


Figure 4.8: XRR curve fit to 204.7 nm SiO₂ thin film

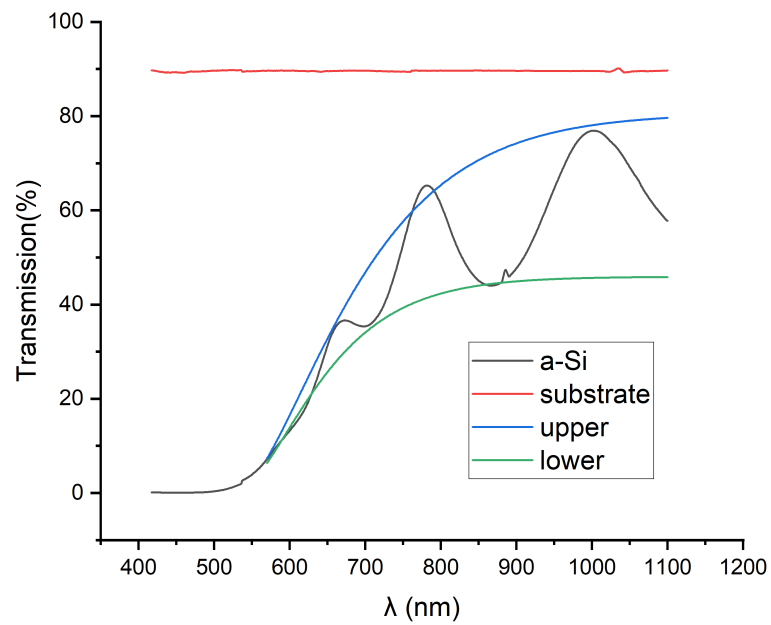


Figure 4.9: The transmission curves for the 600 nm a-Si thin film and the glass substrate together with the upper and lower envelopes for the envelope method

4.2 Refractive index of a-Si thin films

Figure 4.9 shows the UV-Vis transmittance spectra of a 600 nm a-Si thin film deposited on a glass substrate together with the upper and lower envelopes as calculated using the envelope method as described in section 2.5. The equations of the exponential functions fitted to the extrema points listed in Table 4.4 are

$$y = 80.612419 + 1.200883 \times 10^9 e^{-x/89.819343} - 12.00749 \times 10^9 e^{-x/89.821415} \quad (4.3)$$

for the upper envelope with $R^2 = 0.9969$ and

$$y = 45.9178758 + 3.6028012 \times 10^{10} e^{-x/64.3969697} - 3.16015549 \times 10^{10} e^{-x/64.3973202} \quad (4.4)$$

for the lower envelope with $R^2 = 0.9977$. The last d_1 of 281.5 nm is excluded from further calculations due to its distance from the other thickness values. The equation for the refractive index of the a-Si thin film according to Equation 2.35 fitted to the data points of the refractive indices obtained from the envelope method is

$$n^2 = 14.84529 + \frac{7.06709 \times 10^{12} \lambda^2}{\lambda^2 + 2.12449 \times 10^{18}} + \frac{-10.41841 \lambda^2}{\lambda^2 + 190469.027}. \quad (4.5)$$

and the graph of the resulting refractive index versus wavelength is shown in Figure 4.10. The average thickness of the a-Si thin film according to the envelope method is 653 nm.

The refractive index of the a-Si thin film is comparable to the literature values obtained by Swanepoel for a-Si, with values ranging from 3.0 to 3.2 across the relevant wavelengths. However, the refractive index does not decrease monotonically with increasing wavelength as expected for the material. This deviation from the expected behavior is unlikely to be anomalous dispersion and instead is probably caused by the few number of data points from the envelope method used to fit the Sellmeier equation. This is supported by the difference in calculated thickness for the film using the enve-

λ (nm)	T_{film}	T_{sub}	T_M	T_m	n_s	N	n_1	d_1 (nm)	m	d_2 (nm)	n_2
maxima											
1002	76.9	86.3	78.1	45.7	1.62	4.76	3.04	698.1	3.47	659.4	3.07
782	65.3	88.2	62.9	41.4	1.61	4.45	2.93	422.4	4.28	666.8	3.00
655	33.5	89.5	34.3	26.7	1.61	5.34	3.23	564.0	5.64	608.1	3.01
590	11.3	89.7	13.3	11.3	1.61	6.00	3.43				
minima											
871	44.1	86.9	72.3	44.5	1.61	4.59	2.98	737.9	3.91	657.6	3.00
709	36.1	89.6	49.0	35.2	1.61	4.38	2.91	426.7	4.68	670.8	2.99
613	16.4	89.6	21.5	17.7	1.61	5.34	3.23	281.5*			
571	7.4	89.6	7.6	6.6	1.61	8.18	4.02				
								$\bar{d}$	569.8	652.53	

Table 4.4: Relevant values for the envelope method of the a-Si thin film.

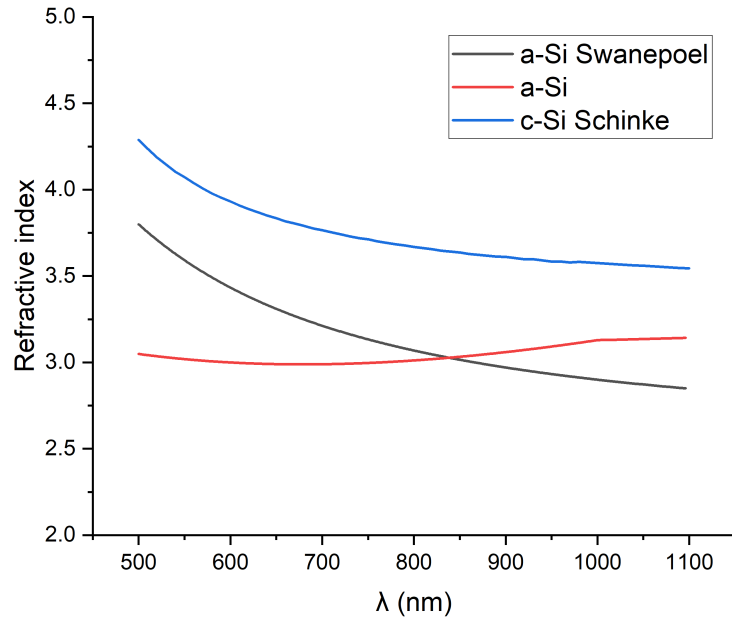


Figure 4.10: Refractive index of a-Si thin film from 500 nm to 1100 nm

lope method and the extrapolation using Equation 4.1. A larger number of oscillations and data points would likely yield a more accurate Sellmeier equation that would exhibit the proper dispersion behavior across the wavelength range. Furthermore, the a-Si thin film refractive index is much lower than that of crystalline silicon, which has a refractive index of 3.5 to 4.25 across the range of wavelengths as measured by Schinke [50]. This is likely due to the lower density and less ordered structure of the a-Si thin film compared to crystalline silicon, and supports the previous finding in section 4.1.

4.3 Refractive index of SiO₂ thin films

Figure 4.11 shows the reflectance spectra of a 990 nm SiO₂ thin film deposited on a Si substrate after deconvolution using a mirror. The extrema points used for the calculation of the refractive index of the film using Equation 2.12 are listed in Table 4.5 together with the corresponding refractive indices and the new wavelengths. Figure 4.12 shows the graph of the refractive index after fitting the data points to Equation 2.35, which is

$$n^2 = 14.84529 + \frac{7.06709 \times 10^{12} \lambda^2}{\lambda^2 + 2.12449 \times 10^{18}} + \frac{-10.41841 \lambda^2}{\lambda^2 + 190469.027}. \quad (4.6)$$

with $R^2 = 0.998$.

The refractive index of the SiO₂ thin film is comparable to the literature values obtained from Gao[51]. The slight increase in refractive index versus the literature values could be attributed to the stoichiometry of the SiO₂ thin film caused by a lack of

λ_0 (nm)	R	λ_1 (nm)	n_1	λ_2 (nm)	n_2
minima					
530.0	0.052	587.5	1.50	676.3	1.50
645.0	0.040	733.8	1.50		
822.5	0.043				
maxima					
577.5	0.154	648.8	1.47	765.0	1.48
720.0	0.140	836.3	1.48		
952.5	0.216				

Table 4.5: Refractive indices and wavelengths for the SiO₂ thin film

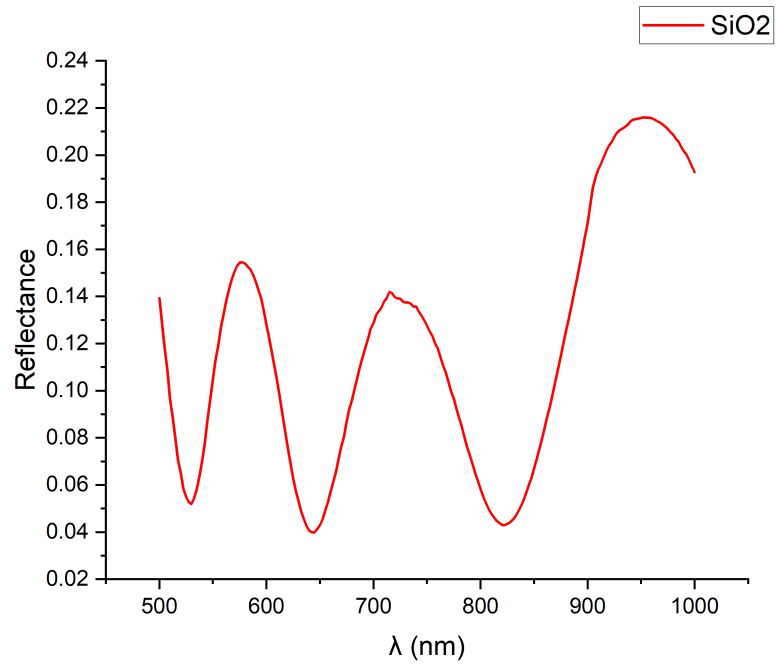


Figure 4.11: Deconvoluted reflectance spectra of 990 nm SiO₂ thin film

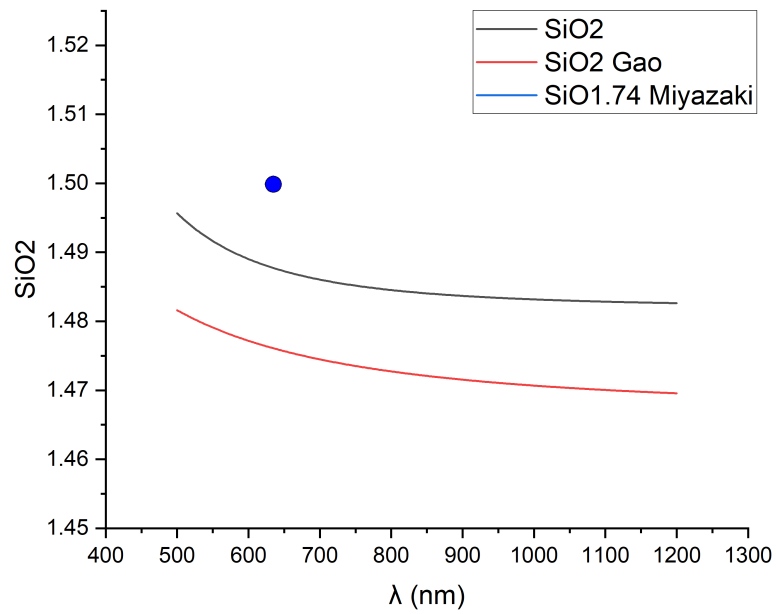


Figure 4.12: Refractive index of the SiO₂ film compared to Gao and Miyazaki

reactive gas during the deposition process, leading to a higher refractive index closer to that of silicon. Miyazaki et al. deposited $\text{SiO}_{1.74}$ thin films and found that the refractive index at 633 nm was 1.50 [52], which is higher than the measured refractive index in this work (1.49). The decreased ratio of oxygen versus silicon may be another reason for the lower density of the SiO_2 thin film as measured by XRR previously compared to bulk SiO_2 aside from the increased porosity of the film.

4.4 Reflectance of a-Si/SiO₂ antireflective coatings

Figure 4.13 shows the reflectance spectra after deconvolution relative to a mirror of a-Si/SiO₂ antireflective coatings with layer thicknesses of 53 nm /14 nm /206 nm, 48 nm /19 nm /208 nm, and 49 nm /24 nm /251 nm for the SiO₂/a-Si/SiO₂ layers together with the simulated reflectance from the TMM using the refractive indices obtained in the previous sections. The simulated reflectance spectra for the first ARC has a value less than 1 % around 656 nm, the simulation of the second ARC has a minimum value less than 1 % at 684 nm, and the third ARC has less than 1 % at 778 nm. The measured reflectance spectra for the ARCs shows the minimum values less than 4 % for the first ARC at 662 nm, 2.3 % for the second ARC at 716 nm, and less than 1 % at 835 nm for the third ARC. The values of the reflectance at 780 nm are around 24 % for the first ARC, 10 % for the second ARC, and 7 % for the third ARC. Table 4.6 summarizes the relevant information of the ARCs being studied. The blank Si wafer exhibits a reflectance ranging from 50 % to 75 % across the wavelength range of 450 nm to 1000 nm.

The multilayer coatings deposited using electron beam deposition have reduced values of reflectance compared to the bare wafer, proving their effectiveness as antireflective coatings. The mismatch in the minimum reflectance values obtained experimentally versus the simulation is due to the nature of the experimental setup, which measures the reflectance of the sample relative to the mirror. More accurate reflectance values would be obtained from absolute measurements which consider the total amount of light incident and reflected from the sample. Interestingly, the deviation of wavelengths between the measured minima of the films and that of their simulated counter parts increases as

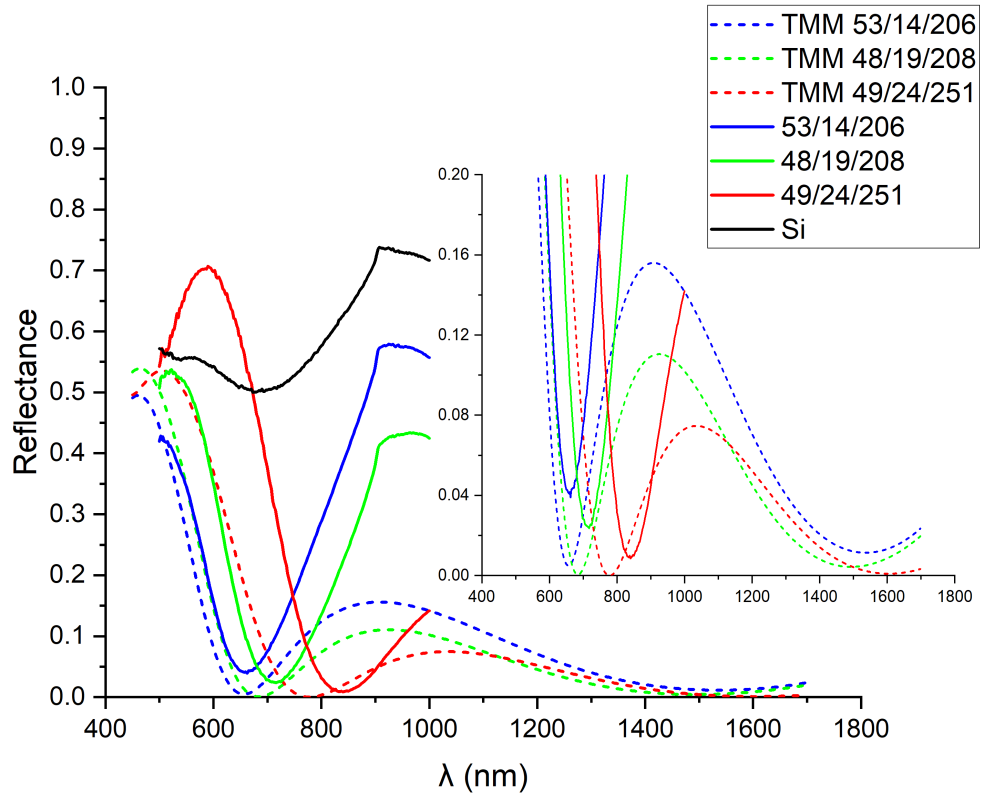


Figure 4.13: Reflectivities of ARCs with varying layer thicknesses simulated using the transfer matrix method and measured experimentally. The reflectance of a bare Si wafer is included for comparison.

	1st minimum		2nd minimum		R at 780 nm
	λ (nm)	R	λ_0 (nm)	R	
53 nm/14 nm/206 nm TMM	656	0.00492	1533	0.0114	0.10945
53 nm/14 nm/206 nm	663	0.03906			0.24194
48 nm/19 nm/208 nm TMM	684	0.00049	1482	0.0043	0.05995
48 nm/19 nm/208 nm	718	0.02367			0.09677
49 nm/24 nm/251 nm TMM	778	0.00003	1606	0.0009	0.00006
49 nm/24 nm/251 nm	840	0.00858			0.06700
Si					0.55323

Table 4.6: Simulated and experimental reflectance of the ARCs and Si wafer at different wavelengths

the target wavelength increases; a difference of 7 nm at a target wavelength of 656 nm, which increases to 34 nm at a target wavelength of 684 nm, and finally 62 nm at a target wavelength of 656 nm. This shift in values is likely caused by the inaccuracy in measuring the refractive index of the thin films which were previously deposited, especially for a-Si. The refractive index values calculated may have been accurate at the lower wavelengths but increasingly erroneous for longer wavelengths, which corroborates the observation in section 4.2 where the index does not monotonically decrease with increasing wavelength.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

In this study, an antireflective coating made of multiple layers of a-Si/SiO₂ with a minimum close to the wavelengths at which terahertz photoconductive antennas are optically pumped has been presented. The transfer matrix model was used to obtain design parameters which supplied information for the thicknesses to be used in the fabrication with electron beam physical vapor deposition. The refractive index data used for the transfer matrix model was obtained from measurements on single films.

The thicknesses for a-Si and SiO₂ obtained using x-ray reflectivity showed that the quartz crystal microbalance readings set at the reference material parameters generally underestimate the actual thicknesses of the deposited films, but follows a linear relationship with them. The densities of the evaporated films are generally lower than that of the bulk. The refractive index of the a-Si film measured via the envelope method compared to the literature shows a deviation from the usual monotonically decreasing value with increasing wavelength. For the refractive index of the SiO₂ film obtained via reflection spectroscopy, the values are higher than the reference, which may be explained by a change in stoichiometric ratio of the film. The measured reflectance of the multilayer coatings generally match the expected minima from the transfer matrix model simulations, although the inaccuracy increases at longer wavelengths.

Recommendations for further studies into the topic include more accurate measurements of the material refractive indices in order to obtain better estimations from the transfer matrix method. Improvements to the quality of the films can also be studied, particularly if substrate heating during growth causes the index to match literature values better. A broader measurement range that includes the infrared region at 1550 nm would also yield more information about the properties of the antireflective coating.

REFERENCES

- [1] L. Yu et al., “The medical application of terahertz technology in non-invasive detection of cells and tissues: Opportunities and challenges,” *RSC Adv.*, vol. 9, no. 17, pp. 9354–9363, Mar. 2019.
- [2] M. Dohi et al., “Application of terahertz pulse imaging as PAT tool for non-destructive evaluation of film-coated tablets under different manufacturing conditions,” *Journal of Pharmaceutical and Biomedical Analysis*, vol. 119, pp. 104–113, Feb. 2016.
- [3] N. V. Petrov, M. S. Kulya, A. N. Tsypkin, V. G. Bespalov, and A. Gorodetsky, “Application of Terahertz Pulse Time-Domain Holography for Phase Imaging,” *IEEE Transactions on Terahertz Science and Technology*, vol. 6, no. 3, pp. 464–472, May 2016.
- [4] L. Song, X. Wu, H. Li, Y. Leng, Y. Tian, and R. Li, “Review of intense terahertz generation and its emerging applications,” *Appl. Phys. Rev.*, vol. 13, no. 2, p. 021 302, Apr. 2026.
- [5] X. Yang, B. Su, Y. Wu, H. Zhang, J. He, and D. R. Jones, “A simple and reliable, optical all-fibre, data acquisition trigger signal for fibre-laser asynchronous optical sampling systems,” *Meas. Sci. Technol.*, vol. 30, no. 4, p. 045 204, Mar. 2019.
- [6] *PCA - Photoconductive antenna*, <https://www.batop.de/products/terahertz/photoconductive-antenna/photoconductive-terahertz-antenna.html>.
- [7] H.-J. Song and T. Nagatsuma, *Handbook of Terahertz Technologies: Devices and Applications*. CRC Press, Apr. 2015, ISBN: 978-981-4613-09-5.
- [8] N. M. Burford and M. O. El-Shenawee, “Review of terahertz photoconductive antenna technology,” *OE*, vol. 56, no. 1, p. 010 901, Jan. 2017.
- [9] L. N. Dela Rosa et al., “Terahertz emission improvement of gallium-arsenide-based bowtie photoconductive antenna by AAO-patterned gold nanoparticles integration on the photoconducting gap,” *Optical Materials*, vol. 150, p. 115 212, Apr. 2024.
- [10] Rituraj, Z. G. Yu, R. M. E. B. Kandegedara, S. Fan, and S. Krishnamurthy, “Highly sensitive and efficient 1550 nm photodetector for room temperature operation,” *AIP Advances*, vol. 15, no. 1, p. 015 215, Jan. 2025.
- [11] X.-Q. Jiang, W.-H. Fan, C. Song, X. Chen, and Q. Wu, “Terahertz photoconductive antenna based on antireflection dielectric metasurfaces with embedded plasmonic nanodisks,” *Appl. Opt., AO*, vol. 60, no. 26, pp. 7921–7928, Sep. 2021.

- [12] S. Alfihed, J. F. Holzman, and I. G. Foulds, “Developments in the integration and application of terahertz spectroscopy with microfluidics,” *Biosensors and Bioelectronics*, vol. 165, p. 112 393, Oct. 2020.
- [13] K. Kawase, Y. Ogawa, Y. Watanabe, and H. Inoue, “Non-destructive terahertz imaging of illicit drugs using spectral fingerprints,” *Opt. Express, OE*, vol. 11, no. 20, pp. 2549–2554, Oct. 2003.
- [14] H. Zhong, A. Redo-Sanchez, and X.-C. Zhang, “Identification and classification of chemicals using terahertz reflective spectroscopic focal-plane imaging system,” *Opt. Express, OE*, vol. 14, no. 20, pp. 9130–9141, Oct. 2006.
- [15] A. Cheng, S. Wu, X. Liu, and H. Lu, “Enhancing concealed object detection in active THz security images with adaptation-YOLO,” *Sci Rep*, vol. 15, no. 1, p. 2735, Jan. 2025.
- [16] F. J. González and G. D. Boreman, “Comparison of dipole, bowtie, spiral and log-periodic IR antennas,” *Infrared Physics & Technology*, vol. 46, no. 5, pp. 418–428, Jun. 2005.
- [17] I. S. Gregory et al., “High resistivity annealed low-temperature GaAs with 100 fs lifetimes,” *Applied Physics Letters*, vol. 83, no. 20, pp. 4199–4201, Nov. 2003.
- [18] R. A. Metzger, A. S. Brown, L. G. McCray, and J. A. Henige, “Structural and electrical properties of low temperature GaInAs,” *Journal of Vacuum Science & Technology B: Microelectronics and Nanometer Structures Processing, Measurement, and Phenomena*, vol. 11, no. 3, pp. 798–801, May 1993.
- [19] M. R. Melloch et al., “Low-Temperature Grown III-V Materials,” *Annual Review of Materials Research*, vol. 25, no. Volume 25, pp. 547–600, Aug. 1995.
- [20] K. Mahalingam, N. Otsuka, M. R. Melloch, J. M. Woodall, and A. C. Warren, “Substrate temperature dependence of arsenic precipitate formation in AlGaAs and GaAs,” *Journal of Vacuum Science & Technology B: Microelectronics and Nanometer Structures Processing, Measurement, and Phenomena*, vol. 9, no. 4, pp. 2328–2332, Jul. 1991.
- [21] K. Rochford, “Polarization and Polarimetry,” in *Encyclopedia of Physical Science and Technology (Third Edition)*, R. A. Meyers, Ed., New York: Academic Press, Jan. 2002, pp. 521–538, ISBN: 978-0-12-227410-7.
- [22] D. A. Lyashenko and Y. P. Svirko, “Crystal Optics,” in *Encyclopedia of Condensed Matter Physics*, F. Bassani, G. L. Liedl, and P. Wyder, Eds., Oxford: Elsevier, Jan. 2005, pp. 283–289, ISBN: 978-0-12-369401-0.
- [23] M. K. Hedayati and M. Elbahri, “Antireflective Coatings: Conventional Stacking Layers and Ultrathin Plasmonic Metasurfaces, A Mini-Review,” *Materials*, vol. 9, no. 6, Jun. 2016.

- [24] D. V. Prashant, S. K. Agnihotri, A. Biswas, and D. P. Samajdar, "Design perspectives of a thin film GaAs solar cell integrated with Carrier Selective contacts and anti-reflection coatings: Optical and device analysis," *Journal of Physics and Chemistry of Solids*, vol. 196, p. 112 396, Jan. 2025.
- [25] D. K. Baisnab, S. Mukherjee, and S. Das, "Chapter 7 - A short review on inorganic thin films from device perspective," in *Chemical Solution Synthesis for Materials Design and Thin Film Device Applications*, S. Das and S. Dhara, Eds., Elsevier, Jan. 2021, pp. 231–275, ISBN: 978-0-12-819718-9.
- [26] S. B. Khan, S. Irfan, and Z. Zhang, "Fabrication and characterization of MgF₂ anti-reflective films comprising dual-layer prepared by physical vapor deposition technique for optoelectronic applications," *Journal of Materials Research and Technology*, vol. 31, pp. 1616–1625, Jul. 2024.
- [27] Y. Jestin, "1.26 - Down-Shifting of the Incident Light for Photovoltaic Applications," in *Comprehensive Renewable Energy*, A. Sayigh, Ed., Oxford: Elsevier, Jan. 2012, pp. 563–585, ISBN: 978-0-08-087873-7.
- [28] A. A. Sharhan, "Transfer Matrix Mathematical Method for Evaluation the DBR Mirror for Light Emitting Diode and Laser," *J. Phys.: Conf. Ser.*, vol. 1535, no. 1, p. 012 018, May 2020.
- [29] S. Jena et al., "Transfer matrix based reverse engineering approach to analyse annealing induced changes in optical transmission of TiO₂/Ag/TiO₂/Ag multilayer," *Optics & Laser Technology*, vol. 144, p. 107 392, Dec. 2021.
- [30] legallyblindbirding, *Anti-Reflection Coatings Part II: The Transfer Matrix Method - Physics, Birds and Blindness*, Dec. 2024.
- [31] M. L. Reed and G. K. Fedder, "2 - Photolithographic Microfabrication," in *Micro Mechanical Systems*, ser. Handbook of Sensors and Actuators, T. Fukuda and W. Menz, Eds., vol. 6, Elsevier Science B.V., Jan. 1998, pp. 13–61.
- [32] J. O'Hanlon, "Gas Properties," in *A User's Guide to Vacuum Technology*, John Wiley & Sons, Ltd, 2003, pp. 9–24, ISBN: 978-0-471-46716-8.
- [33] N. Sharma, T. Saxena, S. N. Alam, B. C. Ray, K. Biswas, and S. K. Jha, "Ceramic-based nanocomposites: A perspective from carbonaceous nanofillers," *Materials Today Communications*, vol. 31, p. 103 764, Jun. 2022.
- [34] S. F. Bent, "Chapter 5 - Semiconductor Surface Chemistry," in *Chemical Bonding at Surfaces and Interfaces*, A. Nilsson, L. G. M. Pettersson, and J. K. Nørskov, Eds., Amsterdam: Elsevier, Jan. 2008, pp. 323–395, ISBN: 978-0-444-52837-7.
- [35] M. Eizenberg, D. A. Smith, M. Heiblum, and A. Segmüller, "Electron beam evaporation of oriented Nb films onto GaAs crystals in ultrahigh vacuum," *Appl. Phys. Lett.*, vol. 49, no. 7, pp. 422–424, Aug. 1986.

- [36] E. Ritter, “Optical film materials and their applications,” *Applied Optics*, Vol. 15, Issue 10, pp. 2318-2327, Oct. 1976.
- [37] T. Xue et al., “Electron beam evaporated SiO₂ interfacial layer for improved CsPbI₃-xBr_x perovskite solar cells,” *Solar Energy Materials and Solar Cells*, vol. 292, p. 113 772, Oct. 2025.
- [38] J. A. Thornton, “Structure-Zone Models Of Thin Films,” in *Modeling of Optical Thin Films*, vol. 0821, SPIE, Feb. 1988, p. 95.
- [39] B. A. Movchan and A. V. Demchishin, “STRUCTURE AND PROPERTIES OF THICK CONDENSATES OF NICKEL, TITANIUM, TUNGSTEN, ALUMINUM OXIDES, AND ZIRCONIUM DIOXIDE IN VACUUM.,” *Fiz. Metal. Metalloved.* 28: 653-60 (Oct 1969)., Jan. 1969.
- [40] H. A. MacLeod and H. A. Macleod, *Thin-Film Optical Filters*, 4th ed. Boca Raton: CRC Press, Mar. 2010, ISBN: 978-0-429-13651-1.
- [41] J. C. Manifacier, J. Gasiot, and J. P. Fillard, “A simple method for the determination of the optical constants n, k and the thickness of a weakly absorbing thin film,” *J. Phys. E: Sci. Instrum.*, vol. 9, no. 11, p. 1002, Nov. 1976.
- [42] R. Swanepoel, “Determination of the thickness and optical constants of amorphous silicon,” *J. Phys. E: Sci. Instrum.*, vol. 16, no. 12, p. 1214, Dec. 1983.
- [43] M. McClain, A. Feldman, D. Kahaner, and X. Ying, “An algorithm and computer program for the calculation of envelope curves,” *Comput. Phys.*, vol. 5, no. 1, pp. 45–48, Jan. 1991.
- [44] M. Ballester et al., “Enhancing the Swanepoel method: Precise envelope detection of thin-film transmission spectra,” *Opt. Express, OE*, vol. 33, no. 6, pp. 13 376–13 400, Mar. 2025.
- [45] E. Márquez et al., “Optical-constant calculation of non-uniform thickness thin films of the Ge₁₀As₁₅Se₇₅ chalcogenide glassy alloy in the sub-band-gap region (0.1–1.8 eV),” *Materials Chemistry and Physics*, vol. 60, no. 3, pp. 231–239, Sep. 1999.
- [46] L. G. Parratt, “Surface Studies of Solids by Total Reflection of X-Rays,” *Phys. Rev.*, vol. 95, no. 2, pp. 359–369, Jul. 1954.
- [47] U. Pietsch, V. Holy, and T. Baumbach, *High-Resolution X-Ray Scattering: From Thin Films to Lateral Nanostructures*. Springer Science & Business Media, Aug. 2004, ISBN: 978-0-387-40092-1.
- [48] H. Kiessig, “Interferenz von Röntgenstrahlen an dünnen Schichten,” *Annalen der Physik*, vol. 402, no. 7, pp. 769–788, 1931.

- [49] L. Nevot and P. Croce, “Characterization of surfaces by grazing X-ray reflection: Application to the study of the polishing of various silicate glasses,” *Rev. Phys. Appl.*, vol. 15, pp. 761–779, 1980.
- [50] C. Schinke et al., “Uncertainty analysis for the coefficient of band-to-band absorption of crystalline silicon,” *AIP Advances*, vol. 5, no. 6, p. 067 168, Jun. 2015.
- [51] W.-S. Liu et al., “Optimal Growth Conditions for Forming c-Axis (002) Aluminum Nitride Thin Films as a Buffer Layer for Hexagonal Gallium Nitride Thin Films Produced with In Situ Continual Radio Frequency Sputtering,” *Micromachines*, vol. 13, no. 9, p. 1546, Sep. 2022.
- [52] H. Miyazaki, “Refractive index and dielectric constant of SiO_x films deposited by reactive sputtering,” *Physics and Chemistry of Glasses - European Journal of Glass Science and Technology Part B*, vol. 51, no. 2, pp. 136–137, Apr. 2010.

Appendices

APPENDIX A

TRANSFER MATRIX METHOD PYTHON CODE

```
# TMM for 9-layers - MJ Hurben Dec 2024
# Modified for 1550 nm and 780 nm optimized SiO2/Si ARC
- GCC 2026

import numpy as np
import matplotlib.pyplot as plt
d2r = np.pi / 180 # Degrees to radians factor
na=1 # Index of air
n_layers=3 # Number of layers

# Define matrices of refractive index n and thickness d,
# modify functions and arrays as needed

# try to use complex refractive index  $n(wl)+k(wl)j$  for
# formula?

def ns(wl): # Define ns as a function of wl based on
# index of substrate material Si from 430 nm
    return 4.0333E-12*wl**4 - 1.7984E-08*wl**3 +
        2.9993E-05*wl**2 - 2.2437E-02*wl + 9.9702

# return -8.974E-15*wl**5 + 5.299E-11*wl**4 -
1.228E-07*wl**3 + 1.398E-04*wl**2 - 7.857E-02*wl +
```

2.103E+01 # In52Ga48As

```
def n1(wl): # Define n1 as a function of wl based on  
refractive index of 1st layer material SiO2  
return np.sqrt(20201856809.3-22607182010.85*wl**2/  
                (wl**2-0.1063836985797) +2405325203.746*wl**2/  
                (wl**2-0.9998803009815))  
  
def n2(wl): # Define n2 as a function of wl a-Si from  
430 nm  
return np.sqrt(14.84529+(7067090000000)*wl**2/  
                (wl**2-(-2124490000000000000)) +(-10.41841)*wl**2/  
                (wl**2-(-190469.027)))  
  
def refractive_index_array(wl): # Add corresponding  
variables and links to functions for each layer  
    n_s = ns(wl)  
    n1_val = n1(wl)  
    n2_val = n2(wl)  
return np.array([na, na, na, na, na, na, n1_val,  
                  n2_val, n1_val, n_s]) # If < 9 layers, use na as  
index of 'missing' layers  
  
d=np.array([1,1,1,1,1,1,1,206,14,53,1000]) # d-Matrix  
values are in nm; initial air layers and final glass  
layer arbitrary  
  
def PS(nva,dva,wl): # Phase Shift calculation
```

```
def M(nA,nB): # Transfer matrix for interface
```

```
    r=(nA-nB)/(nA+nB)
```

```
    t=1+r
```

```
    d=1/t
```

```
    offd=r/t
```

```
    m=np.array([[d, offd],[ offd ,d]])
```

```
    return m
```

```
def P(L,n): # Transfer matrix for propagation
```

```
    L=L*n*2*np.pi
```

```
    v=complex(np.cos(L),-np.sin(L))
```

```
    p=np.array([[v,0],[0,v.conjugate()]])
```

```
    return p
```

```
W=M(nva[8],nva[9]) # Interface of last layer and  
                    glass
```

```
for i in range(8,-1,-1):
```

```
    W=np.matmul(P(dva[i]/wl,nva[i]),W)
```

```
    if i>0:
```

```
        W=np.matmul(M(nva[i-1],nva[i]),W)
```

```
    else:
```

```
        W=np.matmul(M(na,nva[0]),W) # Interface of  
                                     glass and first layer
```

```
    r=W[1,0]/W[0,0]
```

```
    return r
```

```
d_sweep=np.array([1,1,1,1,1,1,206,14,53,1000]) #
```

Matrix for thickness sweep

```
d_optimal=np.array([1,1,1,1,1,1,206,14,53,1000])
```

```
for layer in range(int(9-n_layers),9):    # Rough sweep
    (100 nm range 5 nm step) for best thickness
    original_val=d_sweep[layer]
    base_reflectance= abs( PS(
        refractive_index_array(1550), d_sweep,1550))**2 +
    abs( PS( refractive_index_array(780)
        ,d_sweep,780)) **2
    for sweep_size in range(-50,55,5):
        d_sweep[layer]=original_val+sweep_size
        r_init_1550=
            PS(refractive_index_array(1550),d_sweep,1550)
        r_init_780=
            PS(refractive_index_array(780),d_sweep,780)
        if (abs(r_init_1550)**2+abs(r_init_780)**2)
            <base_reflectance:    # try reducing sum of
                                base_reflectance at 1550 and 780?
            d_optimal[layer]=d_sweep[layer]
            base_reflectance =
                (abs(r_init_1550)**2+abs(r_init_780)**2)
    d_sweep[layer]=d_optimal[layer]
```

```
for layer in range(int(9-n_layers),9):    # Fine sweep
    (10 nm range 1 nm step) for best thickness
    original_val=d_sweep[layer]
    base_reflectance = abs( PS(
        refractive_index_array(1550), d_sweep,1550)) **2+
```

```

    abs( PS( refractive_index_array(780) ,
d_sweep,780)) **2
for sweep_size in range(-5,6):
    d_sweep[layer]=original_val+sweep_size
    r_init_1550 =
        PS(refractive_index_array(1550),d_sweep,1550)
    r_init_780 =
        PS(refractive_index_array(780),d_sweep,780)
    if (abs(r_init_1550)**2+abs(r_init_780)**2)
        <base_reflectance:    # try reducing sum of
base_reflectance at 1550 and 780?
        d_optimal[layer]=d_sweep[layer]
        base_reflectance =
            (abs(r_init_1550)**2+abs(r_init_780)**2)
    d_sweep[layer]=d_optimal[layer]

wls=[] # Array of wavelengths
T=[] # Array of T values
R=[] # Array of R values
R_opt=[] # Array of optimized R values

for wl in range(430,1700,2):
    n=refractive_index_array(wl)
    r=PS(n,d,wl)                # reflectance of initial
    values
    r_opt=PS(n,d_optimal ,wl)    # reflectance of
    optimized values
    wls.append(wl)
    R.append(abs(r)**2)

```

```

        R_opt.append(abs(r_opt)**2)
#        T.append(1-abs(r)**2)

print(d_optimal)
export_array = np.column_stack((wls, R, R_opt))
np.savetxt("tmm.csv", export_array, delimiter=",")

plt.grid(True)
plt.plot(wls, R, label="Initial")
plt.plot(wls, R_opt, label="optimized")
plt.xlim(430, 1700)
plt.ylim(0.00,1)
plt.xlabel('Wavelength■(nm)')
plt.ylabel('Reflectance■R')
plt.legend(loc="upper■right")

```